

Chapter 22

Vector Algebra

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Addition of Vectors and Unit Vectors]

1. What is the maximum number of rectangular components into which a vector can be split in its own plane?
(1) Two (2) Three (3) Four (4) More than four

Sol. Answer (1)

A vector can be split into any number of components. But only two rectangular components in its own plane.

2. Which of the following is a vector?
(1) Electric charge (2) Work (3) Electric current (4) Momentum

Sol. Answer (4)

3. The minimum number of vectors of equal magnitude required to produce zero resultant, is
(1) 2 (2) 3 (3) 4 (4) More than 4

Sol. Answer (1)

Two vectors of equal magnitude and directed in opposite directions give zero resultant.

4. What is the maximum number 'of components into which a vector can split?
(1) Two (2) Three
(3) Four (4) Any number of component

Sol. Answer (4)

5. Which of the following operations between the two vectors can result a vector perpendicular to either of them?
(1) Addition (2) Subtraction (3) Product (4) Division

Sol. Answer (3)

6. If the sum of two unit vector is a unit vector, then the magnitude of their difference is :
(1) 1 (2) $\sqrt{2}$ (3) $\sqrt{5}$ (4) $\sqrt{3}$

Sol. Answer (4)

It is given that $|\vec{a}| = |\vec{b}| = |\vec{a} + \vec{b}| = 1$

$$\Rightarrow OA = AB = OB = 1$$

$\Rightarrow \triangle OAB$ is an equilateral triangle.

$$\text{Since } OA = |\hat{a}| = |-\hat{b}| = AB$$

$\therefore OAB'$ is an isosceles triangle.

$$\angle AB'O = \angle AOB' = 30^\circ$$

$$\angle BOB' = 90^\circ$$

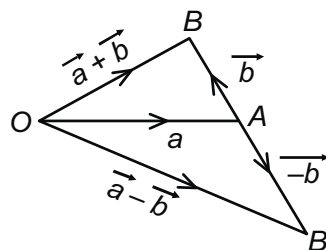
In $\triangle BOB'$, we have

$$BB'^2 = OB^2 + OB'^2$$

$$\Rightarrow 2^2 = |\hat{a} + \hat{b}|^2 + |\hat{a} - \hat{b}|^2$$

$$\Rightarrow 4 = 1 + |\hat{a} - \hat{b}|^2$$

$$\Rightarrow |\hat{a} - \hat{b}| = \sqrt{3}$$



7. Let O be the centre of a regular hexagon $ABCDEF$. Then the magnitude of sum of the vectors $\vec{OA}, \vec{OB}, \vec{OC}, \vec{OD}, \vec{OE}$ and $\vec{OF}$ is

(1) $\sqrt{3}$

(2) 0

(3) 2

(4) 1

Sol. Answer (2)

We know that the centre of a regular hexagon bisects all the diagonals passing through it.

$$\therefore \vec{OA} = -\vec{OD}, \vec{OB} = -\vec{OE} \text{ and } \vec{OC} = -\vec{OF}$$

$$\Rightarrow \vec{OA} + \vec{OD} = 0 \text{ and } \vec{OB} + \vec{OE} = 0$$

$$\text{Hence } \vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} + \vec{OE} + \vec{OF}$$

$$= (\vec{OA} + \vec{OD}) + (\vec{OB} + \vec{OE}) + (\vec{OC} + \vec{OF}) = \vec{0}$$

8. For any two vectors $\vec{a}$ and $\vec{b}$, which one of the following is not true?

(1) $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$

(2) $|\vec{a} - \vec{b}| \leq |\vec{a}| + |\vec{b}|$

(3) $|\vec{a} - \vec{b}| \leq |\vec{a}| - |\vec{b}|$

(4) $|\vec{a} - \vec{b}| \geq |\vec{a}| - |\vec{b}|$

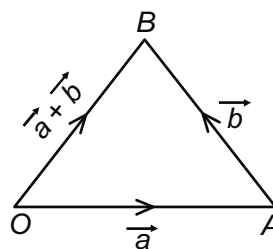
Sol. Answer (3)

Case I : When $\vec{a}$ and $\vec{b}$ are non-collinear. Clearly.

$$OA + AB > OB$$

$$\Rightarrow |\vec{OB}| < |\vec{OA}| + |\vec{AB}|$$

$$\Rightarrow |\vec{a} + \vec{b}| < |\vec{a}| + |\vec{b}|$$

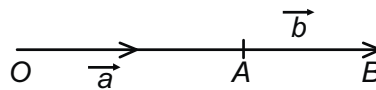


Case II : When $\vec{a}$ and $\vec{b}$ are collinear.

$$\vec{a} + \vec{b} = \vec{OA} + \vec{AB} = \vec{OB}$$

$$\Rightarrow |\vec{OB}| = |\vec{OA}| + |\vec{AB}|$$

$$\Rightarrow |\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$$



Hence in general $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$

Now, $|\vec{a} - \vec{b}| = |\vec{a} + (-\vec{b})| \leq |\vec{a}| + |-\vec{b}|$

$$\Rightarrow |\vec{a} - \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

and $|\vec{a}| = |\vec{a} - \vec{b} + \vec{b}| \leq |\vec{a} - \vec{b}| + |\vec{b}|$

$$\Rightarrow |\vec{a}| - |\vec{b}| \leq |\vec{a} - \vec{b}|$$

$$\Rightarrow |\vec{a} - \vec{b}| \geq |\vec{a}| - |\vec{b}|$$

Hence option (3) is incorrect.

9. The unit vector in the direction of the vector $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, is

(1) $\frac{1}{\sqrt{14}}\hat{i} + \frac{2}{\sqrt{14}}\hat{j} + \frac{3}{\sqrt{14}}\hat{k}$

(2) $\frac{2}{\sqrt{14}}\hat{i} + \frac{1}{\sqrt{14}}\hat{j} + \frac{3}{\sqrt{14}}\hat{k}$

(3) $\frac{1}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{2}{\sqrt{14}}\hat{k}$

(4) $\frac{3}{\sqrt{14}}\hat{i} + \frac{1}{\sqrt{14}}\hat{j} + \frac{2}{\sqrt{14}}\hat{k}$

Sol. Answer (1)

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} + 2\hat{j} + 3\hat{k}}{\sqrt{1^2 + 2^2 + 3^2}} = \left(\frac{1}{\sqrt{14}}\hat{i} + \frac{2}{\sqrt{14}}\hat{j} + \frac{3}{\sqrt{14}}\hat{k} \right)$$

10. The vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ forms a/an

(1) Equilateral triangle

(2) Isosceles triangle

(3) Right triangle

(4) Right Isosceles triangle

Sol. Answer (3)

11. The magnitude of the sum of two vectors is equal to the difference in their magnitudes. What is the angle between vectors?

(1) 0°

(2) 45°

(3) 90°

(4) 180°

Sol. Answer (4)

12. What is the angle between $\vec{a}$ and the resultant of $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$?

- (1) 0 (2) $\tan^{-1} \frac{a}{b}$ (3) $\tan^{-1} \frac{b}{a}$ (4) $\tan^{-1} \frac{(a-b)}{(a+b)}$

Sol. Answer (1)

13. What is the magnitude of the scalar product of the following vectors ?

$$\vec{a} = \hat{i} + \hat{j}; \vec{b} = \hat{i} + \hat{k}$$

- (1) 1 (2) 2 (3) $\sqrt{2}$ (4) 3

Sol. Answer (1)

14. Given that $0.4\hat{i} + 0.8\hat{j} + b\hat{k}$ is a unit vector. What is the value of b ?

- (1) 0.2 (2) $\sqrt{0.2}$ (3) 0.8 (4) $\sqrt{0.8}$

Sol. Answer (2)

15. L and M are the mid points of the sides BC and CD a parallelogram $ABCD$. Then $\vec{AL} + \vec{AM}$ is equal to

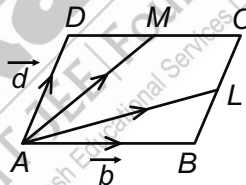
- (1) $\frac{1}{2}\vec{AC}$ (2) $\frac{2}{3}\vec{AC}$ (3) $\frac{3}{2}\vec{AC}$ (4) $\frac{3}{4}\vec{AC}$

Sol. Answer (3)

$$\vec{AL} = \frac{1}{2}(2\vec{b} + \vec{d})$$

$$\vec{AM} = \frac{1}{2}(\vec{b} + 2\vec{d})$$

$$\therefore \vec{AL} + \vec{AM} = \frac{1}{2}(3\vec{b} + 3\vec{d}) = \frac{3}{2}(\vec{b} + \vec{d}) = \frac{3}{2}\vec{AC}$$



16. If $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$, $\vec{c} = \hat{k} + \hat{i}$, a unit vector parallel to $\vec{a} + \vec{b} + \vec{c}$.

- (1) $2\hat{i} + 2\hat{j} + 2\hat{k}$ (2) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ (3) $\frac{(\hat{i} + \hat{j} + \hat{k})}{2\sqrt{8}}$ (4) $\frac{\vec{a} + \vec{b} + \vec{c}}{\sqrt{3}}$

Sol. Answer (2)

17. $p_1\hat{i} + p_2\hat{j}$ is a unit vector perpendicular to $4\hat{i} - 3\hat{j}$ if

- (1) $p_1 = 0.6, p_2 = 0.8$ (2) $p_1 = 3, p_2 = 4$ (3) $p_1 = 0.8, p_2 = 0.6$ (4) $p_1 = 4, p_2 = 3$

Sol. Answer (1)

$$p_1\hat{i} + p_2\hat{j} \text{ is a unit vector} \Rightarrow p_1^2 + p_2^2 = 1$$

$$(p_1\hat{i} + p_2\hat{j}) \perp (4\hat{i} - 3\hat{j}) \Rightarrow (p_1\hat{i} + p_2\hat{j}) \cdot (4\hat{i} - 3\hat{j}) = 0$$

$$\Rightarrow 4p_1 - 3p_2 = 0$$

$$\therefore p_1^2 + \frac{16}{9}p_1^2 = 1 \Rightarrow p_1^2 = \frac{9}{25}$$

$$\Rightarrow \begin{cases} p_1 = 0.6 \\ p_2 = 0.8 \end{cases}$$

18. Let $\vec{a} = 3\hat{i} - \hat{j}$, $\vec{b} = \hat{i} - 2\hat{j}$, $\vec{c} = -\hat{i} + 7\hat{j}$ and $\vec{P} = \vec{a} + \vec{b} + \vec{c}$. Then $\vec{P}$ in terms of $\vec{a}$ and $\vec{b}$ is

- (1) $2\vec{a} + 3\vec{b}$ (2) $-2\vec{a} - 3\vec{b}$ (3) $-2\vec{a} + 3\vec{b}$ (4) $2\vec{a} - 3\vec{b}$

Sol. Answer (4)

$$\vec{p} = \lambda\vec{a} + \mu\vec{b} \Rightarrow 3\hat{i} + 4\hat{j} = \lambda(3\hat{i} - \hat{j}) + \mu(\hat{i} - 2\hat{j})$$

$$\text{or } 3\lambda + \mu = 3, \lambda + 2\mu = -4$$

$$\therefore \lambda = 0, \mu = 3 \text{ i.e., } P = 2\vec{a} - 3\vec{b}$$

19. $ABCDEF$ is a regular hexagon where centre O is the origin, if the position vector of A is $\hat{i} - \hat{j} + 2\hat{k}$, then $\vec{BC}$ is equal to

- (1) $\hat{i} - \hat{j} + 2\hat{k}$ (2) $-\hat{i} + \hat{j} - 2\hat{k}$ (3) $3\hat{i} + 3\hat{j} - 4\hat{k}$ (4) Both (1) & (2)

Sol. Answer (2)

20. The vector $\cos\alpha \cdot \cos\beta \hat{i} + \cos\alpha \cdot \sin\beta \hat{j} + \sin\alpha \hat{k}$ is a/an

- (1) Null vector (2) Unit vector
(3) Constant vector (4) Vector of magnitude 3

Sol. Answer (2)

$$\begin{aligned} |\vec{a}| &= \cos^2\alpha \cdot \cos^2\beta + \cos^2\alpha \sin^2\beta + \sin^2\alpha \\ &= \cos^2\alpha + \sin^2\alpha = 1 \end{aligned}$$

21. If $OACB$ is a parallelogram with $\vec{OC} = \vec{a}$ and $\vec{AB} = \vec{b}$, then $\vec{OA}$ is equal to

- (1) $\vec{a} + \vec{b}$ (2) $\vec{a} - \vec{b}$ (3) $\frac{1}{2}(\vec{b} - \vec{a})$ (4) $\frac{1}{2}(\vec{a} - \vec{b})$

Sol. Answer (4)

22. Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are the position vectors of points A, B, C, D , such that no three of them are collinear and $\vec{a} + \vec{c} = \vec{b} + \vec{d}$, then $ABCD$ is a

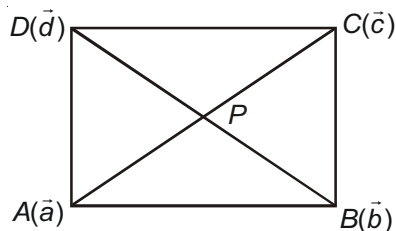
- (1) Rhombus (2) Rectangle (3) Square (4) Parallelogram

Sol. Answer (4)

23. Let $ABCD$ be a parallelogram whose diagonal intersect at P and let O be the origin. Then $\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD}$ equals

- (1) $\vec{OP}$ (2) $2\vec{OP}$ (3) $3\vec{OP}$ (4) $4\vec{OP}$

Sol. Answer (4)



Diagonals of parallelogram intersect at P .

$$\Rightarrow \frac{\vec{a} + \vec{c}}{2} = \vec{P} = \frac{\vec{b} + \vec{d}}{2}$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} + \vec{d} = 2\vec{P} + 2\vec{P} = 4\vec{P}$$

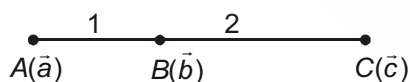
Taking position vectors

$$\vec{OA} + \vec{OB} + \vec{OC} + \vec{OD} = 4\vec{OP}$$

24. $\vec{a}$ and $\vec{b}$ are the position vectors of A, B respectively and C is a point on AB produced such that $AC = 3 AB$. Then the position vector of C is

- (1) $3\vec{b} - 2\vec{a}$ (2) $2\vec{a} - 3\vec{b}$ (3) $2\vec{a} + 3\vec{b}$ (4) $3\vec{a} - 2\vec{b}$

Sol. Answer (1)



$$\vec{b} = \frac{1 \times \vec{c} + 2 \times \vec{a}}{1 + 2}$$

$$\Rightarrow \vec{c} = 3\vec{b} - 2\vec{a}$$

Hence option (1) is correct.

25. The points with position vectors $60\hat{i} + 3\hat{j}$, $40\hat{i} - 8\hat{j}$, $\lambda\hat{i} - 52\hat{j}$ are collinear, if λ is equal to

- (1) -20 (2) 40 (3) -40 (4) 20

Sol. Answer (3)

$$\text{Let } (40\hat{i} - 8\hat{j}) - (60\hat{i} + 3\hat{j}) = \mu [\lambda\hat{i} - 52\hat{j} - 40\hat{i} + 8\hat{j}]$$

$$\Rightarrow \mu\lambda - 40\mu = -20, -44\mu = -11$$

$$\Rightarrow \mu = \frac{1}{4}, \lambda = -40$$

[Product of Vectors, Projection and Volume of Parallelopiped]

26. If $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = 2\vec{a} + 3\vec{b}$, $\overrightarrow{OD} = \vec{a} - 2\vec{b}$, the length of $\overrightarrow{OA}$ is three times the length of $\overrightarrow{OB}$ and $\overrightarrow{OA}$ is perpendicular to $\overrightarrow{DB}$, then $(\overrightarrow{BD} \times \overrightarrow{AC}) \cdot (\overrightarrow{OD} \times \overrightarrow{OC})$ is

- (1) $7|\vec{a} \times \vec{b}|^2$ (2) $42|\vec{a} \times \vec{b}|^2$ (3) 0 (4) 1

Sol. Answer (3)

$$\overrightarrow{BD} \times \overrightarrow{AC} = (\vec{a} - 3\vec{b}) \times (\vec{a} + 3\vec{b}) = 6(\vec{a} \times \vec{b})$$

$$\overrightarrow{OD} \times \overrightarrow{OC} = (\vec{a} - 2\vec{b}) \times (2\vec{a} + 3\vec{b}) = 7(\vec{a} \times \vec{b})$$

$$\therefore (\overrightarrow{BD} \times \overrightarrow{AC}) \cdot (\overrightarrow{OD} \times \overrightarrow{OC}) = 42(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) = 42|\vec{a} \times \vec{b}|^2$$

$$\text{Also } |\vec{a}| = 3|\vec{b}| \text{ and } \overrightarrow{OA} \cdot \overrightarrow{DB} = \vec{a} \cdot (3\vec{b} - \vec{a}) = 0$$

$$\Rightarrow 3\vec{a} \cdot \vec{b} = |\vec{a}| = 9|\vec{b}|^2$$

$$\therefore |\vec{a} \times \vec{b}|^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix} = \begin{vmatrix} 9|\vec{b}|^2 & 3|\vec{b}|^2 \\ 3|\vec{b}|^2 & |\vec{b}|^2 \end{vmatrix} = 0$$

27. Given two vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$, $\vec{b} = -2\hat{i} + 2\hat{j} - \hat{k}$ and $\lambda = \frac{\text{the projection of } \vec{a} \text{ on } \vec{b}}{\text{the projection of } \vec{b} \text{ on } \vec{a}}$, then the value of λ is

- (1) $\frac{3}{7}$ (2) 7 (3) 3 (4) $\frac{7}{3}$

Sol. Answer (4)

$$\lambda = \frac{|\vec{a}| \cos \theta}{|\vec{b}| \cos \theta} = \frac{7}{3}$$

28. The area of parallelogram whose diagonals coincide with the following pair of vectors is $5\sqrt{3}$. The vectors are

- (1) $3\hat{i} + 2\hat{j} - \hat{k}$, $3\hat{i} - \hat{j} + 4\hat{k}$ (2) $\frac{3\hat{i}}{2} - \frac{\hat{j}}{2} - \hat{k}$, $2\hat{i} - 6\hat{j} + 8\hat{k}$
 (3) $3\hat{i} + \hat{j} - 2\hat{k}$, $\hat{i} + 3\hat{j} + 4\hat{k}$ (4) $-(3\hat{i} + \hat{j} - 2\hat{k})$, $\hat{i} - 3\hat{j} - 4\hat{k}$

Sol. Answer (4)

If $\vec{p}$ and $\vec{q}$ denote the diagonals of a parallelogram, then its area is $\frac{1}{2}|\vec{p} \times \vec{q}|$.

Calculate all, finally you will observe that option (4) is correct and rests is wrong.

29. Let $\vec{a}, \vec{b}, \vec{c}$ be three nonzero vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $\lambda \vec{b} \times \vec{a} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$, then λ is

- (1) 1 (2) 2 (3) -1 (4) -2

Sol. Answer (2)

30. If $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$, where $\vec{c} \neq \vec{0}$, then

- (1) $|\vec{a}| = |\vec{c}|, |\vec{b}| = 1$ (2) $|\vec{a}| = |\vec{b}|, |\vec{c}| = 1$ (3) $|\vec{b}| = |\vec{c}|, |\vec{a}| = 1$ (4) $|\vec{a}| \neq |\vec{c}|$

Sol. Answer (1)

31. $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{i} \times \hat{k}) + \hat{k} \cdot (\hat{i} \times \hat{j})$ is equal to

- (1) 3 (2) 0 (3) -3 (4) 1

Sol. Answer (4)

32. If $\vec{a}$ is any vector, then $(\vec{a} \times \hat{i})^2 + (\vec{a} \times \hat{j})^2 + (\vec{a} \times \hat{k})^2$ is equal to

- (1) $|\vec{a}|^2$ (2) $2|\vec{a}|^2$ (3) $3|\vec{a}|^2$ (4) $4|\vec{a}|^2$

Sol. Answer (2)

33. If $|\vec{a} \times \vec{b}| = 2$, $|\vec{a} \cdot \vec{b}| = 2$, then $|\vec{a}|^2 |\vec{b}|^2$ is equal to

- (1) 6 (2) 2 (3) 8 (4) 20

Sol. Answer (3)

34. $|\vec{a} \times \vec{b}|^2$ is equal to

- (1) $|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$ (2) $|\vec{a}|^2 + |\vec{b}|^2 - 3(\vec{a} \cdot \vec{b})^2$ (3) $|\vec{a}|^2 + |\vec{b}|^2 - 2(\vec{a} \cdot \vec{b})$ (4) $|\vec{a}|^2 + |\vec{b}|^2 - \vec{a} \cdot \vec{b}$

Sol. Answer (1)

35. If $\vec{a}$ and $\vec{b}$ are unit vectors, then which of the following values of $\vec{a} \cdot \vec{b}$ is not possible?

- (1) $\sqrt{3}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{-1}{2}$

Sol. Answer (1)

Since $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = \cos \theta$, and $\cos \theta$ never be equal to $\sqrt{3}$.

36. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 5$, $|\vec{c}| = 7$, then angle between $\vec{a}$ and $\vec{b}$ is

- (1) $\frac{\pi}{6}$ (2) $\frac{2\pi}{3}$ (3) $\frac{5\pi}{3}$ (4) $\frac{\pi}{3}$

Sol. Answer (4)

37. If $|\vec{a}| = |\vec{b}|$, then $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})$ is equal to

- (1) Positive (2) Negative (3) Zero (4) None of these

Sol. Answer (3)

38. If $\vec{a}$ and $\vec{b}$ are unit vectors inclined at an angle θ , then the value of $|\vec{a} - \vec{b}|$ is

- (1) $2\sin\frac{\theta}{2}$ (2) $2\sin\theta$ (3) $2\cos\frac{\theta}{2}$ (4) $2\cos\theta$

Sol. Answer (1)

$$\begin{aligned} |\vec{a} - \vec{b}|^2 &= (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b}) \\ &= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\ &= |\vec{a}|^2 - 2(\vec{a} \cdot \vec{b}) + |\vec{b}|^2 \\ &= 2 - 2\cos\theta \quad \left[\begin{array}{l} |\vec{a}|^2 = |\vec{b}|^2 = 1 \\ \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta \\ = \cos\theta \end{array} \right] \\ &= 4\sin^2\frac{\theta}{2} \end{aligned}$$

$$\therefore |\vec{a} - \vec{b}| = 2\sin\frac{\theta}{2}$$

39. The projection of the vector $\hat{i} + \hat{j} + \hat{k}$ along the vector of $\hat{j}$ is

- (1) 1 (2) 0 (3) 2 (4) -1

Sol. Answer (1)

40. If the vectors $3\hat{i} + \lambda\hat{j} + \hat{k}$ and $2\hat{i} - \hat{j} + 8\hat{k}$ are perpendicular, then λ is equal to

- (1) -14 (2) 7 (3) 14 (4) $\frac{1}{7}$

Sol. Answer (3)

41. The vectors $2\hat{i} + 3\hat{j} - 4\hat{k}$ and $a\hat{i} + b\hat{j} + c\hat{k}$ are perpendicular, if

- (1) $a = 2, b = 3, c = -4$ (2) $a = 4, b = 4, c = 5$
(3) $a = 4, b = 4, c = -5$ (4) $a = -4, b = 4, c = -5$

(Hint : Two vectors $\vec{p}$ and $\vec{q}$ are perpendicular if $\vec{p} \cdot \vec{q} = 0$)

Sol. Answer (2)

42. Let $\vec{a}, \vec{b}, \vec{c}$ be three unit vectors such that $|\vec{a} + \vec{b} + \vec{c}| = 1$ and $\vec{a}$ is perpendicular to $\vec{b}$. If $\vec{c}$ makes angle α and β with $\vec{a}$ and $\vec{b}$ respectively, then $\cos\alpha + \cos\beta =$

- (1) $\frac{-3}{2}$ (2) $\frac{3}{2}$ (3) 1 (4) -1

Sol. Answer (4)

43. If θ is the angle between the vectors $2\hat{i} - 2\hat{j} + 4\hat{k}$ and $3\hat{i} + \hat{j} + 2\hat{k}$, then $\sin\theta$ is equal to

- (1) $\frac{2}{3}$ (2) $\frac{2}{\sqrt{7}}$ (3) $\frac{\sqrt{2}}{7}$ (4) $\sqrt{\frac{2}{7}}$

Sol. Answer (2)

44. If $\vec{a} = 2\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 4\hat{j} - 2\hat{k}$, then $\vec{a} \times \vec{b}$ is

- (1) $10\hat{i} + 2\hat{j} + 11\hat{k}$ (2) $10\hat{i} + 3\hat{j} + 11\hat{k}$ (3) $10\hat{i} - 3\hat{j} + 11\hat{k}$ (4) $10\hat{i} - 3\hat{j} - 10\hat{k}$

Sol. Answer (2)

45. If $\vec{a}, \vec{b}$ represent the diagonals of a rhombus, then

- (1) $\vec{a} \times \vec{b} = 0$ (2) $\vec{a} \cdot \vec{b} = 0$ (3) $\vec{a} \cdot \vec{b} = 1$ (4) $\vec{a} \times \vec{b} = \vec{a}$

Sol. Answer (2)

Since diagonals of a rhombus are perpendicular to each other then $\vec{a} \cdot \vec{b} = 0$.

46. If $\vec{u} = \vec{a} - \vec{b}$, $\vec{v} = \vec{a} + \vec{b}$ and $|\vec{a}| = |\vec{b}| = 2$, then $|\vec{u} \times \vec{v}|$ is equal to

- (1) $2\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$ (2) $\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$ (3) $2\sqrt{4 - (\vec{a} \cdot \vec{b})^2}$ (4) $[4 - (\vec{a} \cdot \vec{b})^2]$

Sol. Answer (1)

47. A vector $\vec{c}$ of magnitude $\sqrt{7}$ which is perpendicular to the vector $\vec{a} = 2\hat{j} - \hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} - 3\hat{k}$ and makes an obtuse angle with the y-axis is given by

- (1) $\left(\frac{-4}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-2}{\sqrt{3}}\right)$ (2) $\left(\frac{-4}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$
 (3) $\left(\frac{4}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-2}{\sqrt{3}}\right)$ (4) Both (2) & (3)

Sol. Answer (3)

48. If $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{i} - \hat{j}$, then $\vec{a} \cdot \vec{b} =$

- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{4}{3}$ (4) 0

Sol. Answer (4)

49. Let $\vec{a}$ and $\vec{b}$ are unit vectors inclined at an angle α to each other, if $|\vec{a} + \vec{b}| < 1$ then

- (1) $\alpha = \frac{\pi}{2}$ (2) $\alpha < \frac{\pi}{3}$ (3) $\alpha > \frac{2\pi}{3}$ (4) $\frac{\pi}{3} < \alpha < \frac{2\pi}{3}$

Sol. Answer (3)

$$|\vec{a} + \vec{b}| < 1$$

$$\Rightarrow |\vec{a} + \vec{b}|^2 < 1$$

$$\Rightarrow a^2 + b^2 + 2\vec{a} \cdot \vec{b} < 1$$

$$\Rightarrow 1 + 1 + 2 \cos \alpha < 1$$

$$2 \cos \alpha < -1$$

$$\cos \alpha < -\frac{1}{2}$$

$$\alpha > \frac{2\pi}{3}$$

50. A unit vector in the xy-plane that makes an angle of $\frac{\pi}{4}$ with the vector $\hat{i} + \hat{j}$ and an angle of $\frac{\pi}{3}$ with the vector

$$3\hat{i} - 4\hat{j} \text{ is}$$

(1) $\hat{i}$

(2) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$

(3) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

(4) Not in existence

Sol. Answer (4)

Let $\vec{a}$ is the required vector

$$\vec{a} = x\hat{i} + y\hat{j}$$

... (i) where $x^2 + y^2 = 1$

Let $\vec{b} = \hat{i} + \hat{j}$

... (ii)

$$\vec{c} = 3\hat{i} - 4\hat{j}$$

... (iii)

$$x^2 + y^2 = 1$$

... (iv)

Angle between $\vec{a}$ and $\vec{b} = \frac{\pi}{4}$

$$\cos \frac{\pi}{4} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \Rightarrow \frac{1}{\sqrt{2}} = \frac{x+y}{\sqrt{x^2+y^2} \sqrt{1+1}} = \frac{x+y}{\sqrt{2}}$$

$$\Rightarrow x+y=1 \quad \dots (v)$$

Again angle between $\vec{a}$ and $\vec{c}$ is $\frac{\pi}{3}$

$$\Rightarrow \cos \frac{\pi}{3} = \frac{\vec{a} \cdot \vec{c}}{|\vec{a}| |\vec{c}|} = \frac{3x-4y}{\sqrt{x^2+y^2} \sqrt{9+16}} = \frac{3x-4y}{5}$$

$$\Rightarrow \frac{1}{2} = \frac{3x-4y}{5}$$

$$3x - 4y = \frac{5}{2} \quad \dots (vi)$$

By (v) and (vi)

$$3(x) - 4(1-x) = \frac{5}{2}$$

$$\Rightarrow 3x - 4 + 4x = \frac{5}{2}$$

$$\Rightarrow -4 + 7x = \frac{5}{2}$$

$$\Rightarrow 7x = \frac{5}{2} + 4 = \frac{13}{2}$$

$$\Rightarrow x = \frac{13}{14} \quad \dots (vii)$$

$$\text{Similarly } y = \frac{1}{14} \quad \dots (viii)$$

$$\text{But } x^2 + y^2 = \left(\frac{13}{14}\right)^2 + \left(\frac{1}{14}\right)^2 = \frac{170}{196} \neq 1$$

Hence no such vector exists.

51. The vectors $2\hat{i} - m\hat{j} + 3m\hat{k}$ and $(1+m)\hat{i} - 2m\hat{j} + \hat{k}$ include an acute angle for

(1) All real m

(2) $m < -2$ or $m > -\frac{1}{2}$

(3) $m = -\frac{1}{2}$

(4) $m \in \left[-2, -\frac{1}{2}\right]$

Sol. Answer (2)

For acute angle $\vec{a} \cdot \vec{b} > 0$

$$\Rightarrow 2(1+m) + 2m^2 + 3m > 0$$

$$\Rightarrow 2m^2 + 5m + 2 > 0$$

$$\Rightarrow 2m^2 + 4m + m + 2 > 0$$

$$\Rightarrow 2m(m+2) + 1(m+2) > 0$$

$$\Rightarrow (2m+1)(m+2) > 0$$

$$\Rightarrow m \in (-\infty, -2) \cup \left(-\frac{1}{2}, \infty\right)$$

52. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors such that $|\vec{a}| = 3$, $|\vec{b}| = 3$, $|\vec{c}| = 2$, $|\vec{a} + \vec{b} + \vec{c}| = 4$ and $\vec{a}$ is perpendicular to $\vec{b}, \vec{c}$ makes angle θ and ϕ with $\vec{a}$ and $\vec{b}$ respectively, then $\cos \theta + \cos \phi =$
- (1) $\frac{3}{4}$ (2) $-\frac{3}{4}$ (3) $\frac{1}{2}$ (4) $-\frac{1}{2}$

Sol. Answer (4)

$$|\vec{a} + \vec{b} + \vec{c}|^2 = 16$$

$$a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 16$$

$$9 + 9 + 4 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 16$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -3 \quad \dots (i)$$

We have, $\vec{a} \cdot \vec{b} = 0$

$$\vec{b} \cdot \vec{c} = |\vec{b}| |\vec{c}| \cos \phi = 6 \cos \phi$$

$$\vec{c} \cdot \vec{a} = |\vec{c}| |\vec{a}| \cos \theta = 6 \cos \theta$$

By (i)

$$0 + 6(\cos \theta + \cos \phi) = -3$$

$$\cos \theta + \cos \phi = -\frac{1}{2}$$

53. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ be three vectors. A vector in the plane of $\vec{b}$ and $\vec{c}$ whose projection on $\vec{a}$ is $\frac{\sqrt{2}}{3}$ is

- (1) $-(2\hat{i} + 3\hat{j} - 3\hat{k})$ (2) $2\hat{i} - 3\hat{j} + 6\hat{k}$ (3) $\hat{i} - \hat{j} + 2\hat{k}$ (4) $3\hat{i} + 2\hat{j} - \hat{k}$

Sol. Answer (1)

Let vector in the plane of $\vec{b}$ and $\vec{c}$ is $\vec{d}$

$$\vec{d} = x(\hat{i} + 2\hat{j} - \hat{k}) + y(\hat{i} + \hat{j} - 2\hat{k}) \quad \dots (i)$$

$$\frac{\vec{a} \cdot \vec{d}}{|\vec{a}|} = \frac{\sqrt{2}}{3}$$

$$\Rightarrow \frac{(2\hat{i} - \hat{j} + \hat{k}) \cdot (x(\hat{i} + 2\hat{j} - \hat{k}) + y(\hat{i} + \hat{j} - 2\hat{k}))}{\sqrt{6}} = \frac{\sqrt{2}}{3}$$

$$\Rightarrow 2(x + y) - (2x + y) + (-x - 2y) = 2$$

$$\Rightarrow 2x + 2y - 2x - y - x - 2y = 2$$

$$\Rightarrow x + y = -2$$

54. The area of a parallelogram whose diagonals are given by the vectors $3\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + 4\hat{k}$ is

(1) $10\sqrt{3}$

(2) $5\sqrt{3}$

(3) 8

(4) 4

Sol. Answer (2)

$$\text{Area of parallelogram} = \left| \frac{1}{2} \vec{d}_1 \times \vec{d}_2 \right|$$

Where $\vec{d}_1$ and $\vec{d}_2$ are the diagonal

$$\begin{aligned} \frac{1}{2} \vec{d}_1 \times \vec{d}_2 &= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} \\ &= \frac{1}{2} [\hat{i}(4-6) - \hat{j}(12+2) + \hat{k}(-9-1)] \\ &= \frac{1}{2} [-2\hat{i} - 14\hat{j} - 10\hat{k}] \\ &= -\hat{i} - 7\hat{j} - 5\hat{k} \end{aligned}$$

$$A = \left| \frac{1}{2} \vec{d}_1 \times \vec{d}_2 \right| = \sqrt{1+49+25} = 5\sqrt{3}$$

55. If $\vec{a}$ and $\vec{b}$ are two unit vectors, then the vector $(\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})$ is parallel to the vector

(1) $\vec{a} - \vec{b}$

(2) $\vec{a} + \vec{b}$

(3) $2\vec{a} - \vec{b}$

(4) $2\vec{a} + \vec{b}$

Sol. Answer (1)

$$\text{Let } \vec{x} = (\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})$$

$$\begin{aligned} &= \vec{a} \times (\vec{a} \times \vec{b}) + \vec{b} \times (\vec{a} \times \vec{b}) \\ &= \{(\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b}\} + \{(\vec{b} \cdot \vec{b})\vec{a} - (\vec{b} \cdot \vec{a})\vec{b}\} \\ &= (\vec{a} \cdot \vec{b})(\vec{a} - \vec{b}) + 1(\vec{a} - \vec{b}) \quad (\because |\vec{a}| = |\vec{b}| = 1) \end{aligned}$$

$$\vec{x} = (\vec{a} - \vec{b})(\vec{a} \cdot \vec{b} + 1)$$

Hence $\vec{x}$ is parallel to $\vec{a} - \vec{b}$.

56. Let $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c}$ is a vector such that $|\vec{a} \times \vec{c}|^2 + (\vec{a} \cdot \vec{c})^2 = 144$ then $|\vec{c}|$ is equal to

(1) 16

(2) 4

(3) 3

(4) 9

Sol. Answer (2)

57. $ABCD$ is a quadrilateral with $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AD} = \vec{b}$ and $\overrightarrow{AC} = 2\vec{a} + 3\vec{b}$. If its area is α times the area of the parallelogram with AB and AD as adjacent sides, then $\alpha =$

- (1) 5 (2) $\frac{5}{2}$ (3) 1 (4) $\frac{1}{2}$

Sol. Answer (2)

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

$$\Rightarrow \overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB} = 2\vec{a} + 3\vec{b} - \vec{a} = \vec{a} + 3\vec{b}$$

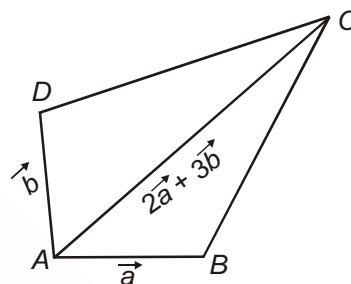
$$\text{Area of quadrilateral } ABCD = \alpha(\vec{a} \times \vec{b})$$

$$\Rightarrow \frac{1}{2}(\vec{a} \times (2\vec{a} + 3\vec{b})) + \frac{1}{2}[(2\vec{a} + 3\vec{b}) \times \vec{b}] = \alpha(\vec{a} \times \vec{b})$$

$$\Rightarrow \frac{1}{2}[3(\vec{a} \times \vec{b})] + \frac{1}{2}[2(\vec{a} \times \vec{b})] = \alpha(\vec{a} \times \vec{b})$$

$$\Rightarrow \frac{3}{2}(\vec{a} \times \vec{b}) + (\vec{a} \times \vec{b}) = \alpha(\vec{a} \times \vec{b})$$

$$\Rightarrow \alpha = \frac{5}{2}$$



58. A unit vector perpendicular to the plane passing through the points whose position vectors are $\hat{i} - \hat{j} + 2\hat{k}$, $2\hat{i} - \hat{k}$ and $2\hat{i} + \hat{k}$ is

- (1) $2\hat{i} + \hat{j} + \hat{k}$ (2) $\frac{1}{\sqrt{6}}(\hat{i} + 2\hat{j} + \hat{k})$
 (3) $\pm \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}} \right)$ (4) $\frac{1}{\sqrt{6}}(3\hat{i} + 4\hat{j} - \hat{k})$

Sol. Answer (3)

$$\vec{a} = (2\hat{i} - \hat{k}) - (\hat{i} - \hat{j} + 2\hat{k}) = \hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{b} = (2\hat{i} + \hat{k}) - (2\hat{i} - \hat{k}) = 2\hat{k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -3 \\ 0 & 0 & 2 \end{vmatrix} = \hat{i}(2) - \hat{j}(2) + \hat{k}(0) = 2\hat{i} - 2\hat{j}$$

$$\text{Unit vector} = \pm \frac{2\hat{i} - 2\hat{j}}{|2\hat{i} - 2\hat{j}|} = \pm \frac{\hat{i} - \hat{j}}{\sqrt{2}}$$

59. Let $\vec{a}, \vec{b}, \vec{c}$ be the position vectors of three vertices A, B, C of a triangle respectively. Then the area of the triangle is

(1) $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$

(2) $\frac{1}{2}(\vec{a} \times \vec{b}) \cdot \vec{c}$

(3) $\frac{1}{2}|\vec{a} \times \vec{b}|$

(4) $\frac{1}{2}|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$

Sol. Answer (4)

$$\vec{AB} = \vec{b} - \vec{a}$$

$$\vec{BC} = \vec{c} - \vec{b}$$

$$\begin{aligned} \text{Area of triangle} &= \frac{1}{2} |\vec{AB} \times \vec{BC}| \\ &= \frac{1}{2} |(\vec{b} - \vec{a}) \times (\vec{c} - \vec{b})| \\ &= \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}| \end{aligned}$$

60. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a}$ is perpendicular to the plane $\vec{b}$ and $\vec{c}$ and angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$,

then value of $|\vec{a} + \vec{b} + \vec{c}|$ is

(1) 4

(2) 2

(3) 9

(4) 3

Sol. Answer (2)

$$\begin{aligned} |\vec{a} + \vec{b} + \vec{c}|^2 &= (\vec{a} + \vec{b} + \vec{c})(\vec{a} + \vec{b} + \vec{c}) \\ &= |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2[\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}] \\ &= 1 + 1 + 1 + 2\left(0 + \cos \frac{\pi}{3} + 0\right) \\ &= 4 \end{aligned}$$

$$\therefore |\vec{a} + \vec{b} + \vec{c}| = 2$$

61. If $\vec{a} \cdot \hat{i} = \vec{a} \cdot (\hat{i} + \hat{j}) = \vec{a} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$, then $\vec{a} =$

(1) Zero vector

(2) $\hat{i}$

(3) $\hat{j}$

(4) $\hat{i} + \hat{j} + \hat{k}$

Sol. Answer (2)

$$\vec{a} \cdot \hat{i} = 1$$

$$(\text{Let } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k})$$

$$\Rightarrow a_1 = 1$$

...(i)

$$\vec{a} \cdot (\hat{i} + \hat{j}) = 1$$

$$\Rightarrow a_1 + a_2 = 1 \Rightarrow a_2 = 0$$

$$\text{Similarly } a_3 = 0$$

$$\therefore \vec{a} = 1 \cdot \hat{i} + 0 \cdot \hat{j} + 0 \cdot \hat{k}$$

$$\Rightarrow \vec{a} = \hat{i}$$

62. The vectors $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = -2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\vec{c} = \hat{i} - 3\hat{j} + \lambda\hat{k}$ are coplanar if the value of λ is

(1) 1

(2) 2

(3) 3

(4) 5

Sol. Answer (4)

Three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar if $[\vec{a} \vec{b} \vec{c}] = 0$

$$\therefore \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & \lambda \end{vmatrix} = 0 \Rightarrow -\lambda + 5 = 0 \Rightarrow \lambda = 5$$

63. The volume of the parallelepiped whose edges are $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$ is

(1) -2 cubic unit

(2) 2 cubic unit

(3) 1 cubic unit

(4) 4 cubic unit

Sol. Answer (2)

Volume of parallelepiped = $[\vec{a} \vec{b} \vec{c}]$

$$= \begin{vmatrix} 2 & -3 & 4 \\ 1 & 2 & -1 \\ 2 & -1 & 2 \end{vmatrix} = 2(4 - 1) + 3(2 + 2) + (-1 - 4)$$

$$= 6 + 12 - 20 = -2$$

As volume is always +ve, therefore the required volume is 2 cubic cm.

64. Let a, b, c be three distinct positive real numbers. If $\vec{p}, \vec{q}, \vec{r}$ lie in a plane, where $\vec{p} = a\hat{i} - a\hat{j} + b\hat{k}$, $\vec{q} = \hat{i} + \hat{k}$

and $\vec{r} = c\hat{i} + c\hat{j} + b\hat{k}$, then b is

(1) The A.M. of a, c (2) The G.M. of a, c (3) The H.M. of a, c

(4) Equal to 0

Sol. Answer (3)

$$\begin{vmatrix} a & -a & b \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

$$\Rightarrow a(0 - c) + a(b - c) + b(c) = 0$$

$$\Rightarrow -ac + ab - ac + bc = 0$$

$$\Rightarrow b = \frac{2ac}{a + c}$$

Hence, a, b, c are in H.P.

65. If $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $[\vec{a} \ \vec{b} \ \vec{c}] = 4$, then $[\vec{a} \times \vec{b} \ \vec{b} \times \vec{c} \ \vec{c} \times \vec{a}]$ is equal to

- (1) 8 (2) 64 (3) 4 (4) 16

Sol. Answer (4)

$$\begin{aligned} & [\vec{a} \times \vec{b} \ \vec{b} \times \vec{c} \ \vec{c} \times \vec{a}] \\ &= (\vec{a} \times \vec{b}) \cdot [(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})] \\ &= (\vec{a} \times \vec{b}) \cdot [((\vec{b} \times \vec{c}) \cdot \vec{a})\vec{c} - ((\vec{b} \times \vec{c}) \cdot \vec{c})\vec{a}] \\ &= [\vec{a}\vec{b}\vec{c}]^2 = 4^2 = 16 \end{aligned}$$

[Solving of Vector Equations]

66. If $\vec{r} = x(\vec{a} \times \vec{b}) + y(\vec{b} \times \vec{c}) + z(\vec{c} \times \vec{a})$ and $[\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{3}$, then $x + y + z$ is equal to

- (1) $7\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$ (2) $\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$ (3) $5\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$ (4) $3\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$

Sol. Answer (4)

$$\vec{r} = x(\vec{a} \times \vec{b}) + y(\vec{b} \times \vec{c}) + z(\vec{c} \times \vec{a})$$

Dot product by $\vec{c}$

$$\vec{r} \cdot \vec{c} = x[\vec{a}\vec{b}\vec{c}] + 0 + 0$$

$$x = \frac{\vec{r} \cdot \vec{c}}{[\vec{a}\vec{b}\vec{c}]} = 3\vec{r} \cdot \vec{c} \quad \left(\because [\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{3} \right)$$

$$\text{Similarly } y = 3\vec{r} \cdot \vec{a}$$

$$\text{and } z = 3\vec{r} \cdot \vec{b}$$

$$x + y + z = 3\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c})$$

67. If the vertices of a tetrahedron have the position vectors $\vec{0}, \hat{i} + \hat{j}, 2\hat{j} - \hat{k}$ and $\hat{i} + \hat{k}$ then the volume of the tetrahedron is

- (1) $\frac{1}{6}$ (2) 1 (3) 2 (4) 3

Sol. Answer (1)

Edges of the tetrahedron are

$$\vec{a} = \hat{i} + \hat{j} - \vec{0} = \hat{i} + \hat{j}$$

$$\text{Similarly } \vec{b} = (2\hat{j} - \hat{k}) - \vec{0} = 2\hat{j} - \hat{k}$$

$$\vec{c} = (\hat{i} + \hat{k}) - \vec{0} = (\hat{i} + \hat{k})$$

Volume of tetrahedron

$$= \frac{1}{6} [\vec{a}\vec{b}\vec{c}] = \frac{1}{6} \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & -1 \\ 1 & 0 & 1 \end{vmatrix} = \frac{1}{6} [1(2) - 1(1)] = \frac{1}{6}$$

68. If $[(2\vec{a} + \vec{b}) \vec{c} \vec{d}] = \lambda [\vec{a} \vec{c} \vec{d}] + \mu [\vec{b} \vec{c} \vec{d}]$ then $\lambda + \mu$ is equal to

- (1) -6 (2) 3 (3) 2 (4) -2

Sol. Answer (2)

$$[2\vec{a} + \vec{b} \vec{c} \vec{d}] = (2\vec{a} + \vec{b}) \cdot (\vec{c} \times \vec{d})$$

$$= 2\vec{a} \cdot (\vec{c} \times \vec{d}) + \vec{b} \cdot (\vec{c} \times \vec{d})$$

$$= 2[\vec{a} \vec{c} \vec{d}] + [\vec{b} \vec{c} \vec{d}] = \lambda[\vec{a} \vec{c} \vec{d}] + \mu[\vec{b} \vec{c} \vec{d}]$$

$$\Rightarrow \lambda = 2, \mu = 1$$

$$\lambda + \mu = 2 + 1 = 3$$

69. If $\vec{p} = 2\vec{a} - 3\vec{b}$, $\vec{q} = \vec{a} - 2\vec{b} + \vec{c}$, $\vec{r} = -3\vec{a} + \vec{b} + 2\vec{c}$, $\vec{a}$, $\vec{b}$, $\vec{c}$ being non-zero, non-coplanar vectors, then $-2\vec{a} + \vec{b} - \vec{c}$ is equal to

- (1) $\vec{p} - 4\vec{q}$ (2) $\frac{(-7\vec{q} + \vec{r})}{5}$ (3) $2\vec{p} - 3\vec{q} + \vec{r}$ (4) $4\vec{p} - 2\vec{r}$

Sol. Answer (3)

70. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-coplanar vectors such that $\vec{a} + \vec{b} + \vec{c} = \alpha\vec{d}$ and $\vec{b} + \vec{c} + \vec{d} = \beta\vec{a}$ then $\vec{a} + \vec{b} + \vec{c} + \vec{d}$ is equal to

- (1) $\alpha\vec{a}$ (2) $\beta\vec{b}$ (3) 0 (4) $(\alpha + \beta)\vec{c}$

Sol. Answer (3)

$$\vec{a} + \vec{b} + \vec{c} = \alpha\vec{d} \quad \dots (i)$$

$$\vec{b} + \vec{c} + \vec{d} = \beta\vec{a} \quad \dots (ii)$$

By (i) - (ii)

$$\vec{a} - \vec{d} = \alpha\vec{d} - \beta\vec{a}$$

$$\Rightarrow \beta = -1, \alpha = -1$$

$$\text{Or } \vec{a} + \vec{b} + \vec{c} = -\vec{d}$$

$$\vec{a} + \vec{b} + \vec{c} + \vec{d} = \vec{0}$$

71. The position vector of three points are $2\vec{a} - \vec{b} + 3\vec{c}$, $\vec{a} - 2\vec{b} + \lambda\vec{c}$ and $\mu\vec{a} - 5\vec{b}$ where $\vec{a}$, $\vec{b}$, $\vec{c}$ are non coplanar vectors. The points are collinear when

- (1) $\lambda = -2, \mu = \frac{9}{4}$ (2) $\lambda = \frac{-9}{4}, \mu = 2$ (3) $\lambda = \frac{9}{4}, \mu = -2$ (4) $\lambda = 2, \mu = -2$

Sol. Answer (3)

$$(\vec{a} - 2\vec{b} + \lambda\vec{c}) - (2\vec{a} - \vec{b} + 3\vec{c}) = t((\mu\vec{a} - 5\vec{b}) - (\vec{a} - 2\vec{b} + \lambda\vec{c}))$$

Comparing both the sides, we get

$$-1 = t(\mu - 1) \quad \dots(i)$$

$$-1 = t(-5 + 2) \quad \dots(ii)$$

$$\lambda - 3 = -\lambda t \quad \dots(iii)$$

From (ii)

$$t = \frac{1}{3}$$

From (iii)

$$\lambda - 3 = -\frac{\lambda}{3}$$

$$\frac{4\lambda}{3} = 3$$

$$\text{From (i)} \quad -1 = \frac{1}{3}(\mu - 1)$$

$$\mu - 1 = -3$$

$$\mu = -2$$

$$\boxed{\lambda = \frac{9}{4}, \mu = -2}$$

72. If $\vec{p} \times \vec{q} = \vec{r}$ and $\vec{p} \cdot \vec{q} = c$, then $\vec{q}$ is

$$(1) \quad \frac{c\vec{p} - \vec{p} \times \vec{r}}{|\vec{p}|^2}$$

$$(2) \quad \frac{c\vec{p} + \vec{p} \times \vec{r}}{|\vec{p}|^2}$$

$$(3) \quad \frac{c\vec{r} - \vec{p} \times \vec{r}}{|\vec{p}|^2}$$

$$(4) \quad \frac{c\vec{r} + \vec{p} \times \vec{r}}{|\vec{p}|^2}$$

Sol. Answer (1)

$$(\vec{p} \times \vec{q}) = \vec{r}$$

Applying cross product by $\vec{p}$

$$\vec{p} \times (\vec{p} \times \vec{q}) = \vec{p} \times \vec{r}$$

$$\Rightarrow (\vec{p} \cdot \vec{q})\vec{p} - (\vec{p} \cdot \vec{p})\vec{q} = \vec{p} \times \vec{r}$$

$$\Rightarrow c\vec{p} - p^2\vec{q} = \vec{p} \times \vec{r}$$

$$\Rightarrow \vec{q} = \frac{c\vec{p} - \vec{p} \times \vec{r}}{|\vec{p}|^2}$$

73. If $\vec{a} = (\hat{i} + \hat{j} + \hat{k})$, $\vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$. Then $\vec{b}$ is

$$(1) \quad \hat{i} - \hat{j} + \hat{k}$$

$$(2) \quad 2\hat{j} - \hat{k}$$

$$(3) \quad \hat{i}$$

$$(4) \quad -\hat{i} + 2\hat{j}$$

Sol. Answer (3)

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{a} \cdot \vec{b} = 1, \vec{a} \times \vec{b} = \hat{j} - \hat{k}$$

$$\vec{a} \times (\vec{a} \times \vec{b}) = \vec{a} \times (\hat{j} - \hat{k})$$

$$(\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} = \vec{a} \times (\hat{j} - \hat{k}) \quad \dots (i)$$

$$\text{as } \vec{a} \cdot \vec{b} = 1$$

$$\vec{a} \cdot \vec{a} = a^2 = 1+1+1=3$$

$$\vec{a} \times (\hat{j} - \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= \hat{i}(-1-1) - \hat{j}(-1) + \hat{k}(1-0)$$

$$= -2\hat{i} + \hat{j} + \hat{k}$$

Putting all value in equation (i)

$$(1)(\hat{i} + \hat{j} + \hat{k}) - 3\vec{b} = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow 3\vec{b} = 3\hat{i} \Rightarrow \vec{b} = \hat{i}$$

74. The position vectors of the vertices of triangle are $3\hat{i} + 4\hat{j} + 5\hat{k}$, $\hat{i} + 7\hat{k}$ and $5\hat{i} + 5\hat{j}$. The distance between the circumcentre and the ortho centre is

- (1) 0 (2) $\frac{3\sqrt{274}}{\sqrt{11}}$ (3) $\sqrt{306}$ (4) $\frac{3}{2}\sqrt{306}$

Sol. Answer (2)Let ABC is the given triangle and the position vector of A, B and C are $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively

$$\overline{AB} = \vec{b} - \vec{a} = (\hat{i} + 7\hat{k}) - (3\hat{i} + 4\hat{j} + 5\hat{k}) = -2\hat{i} - 4\hat{j} + 2\hat{k}$$

$$\Rightarrow AB = \sqrt{24} = c$$

$$\text{Similarly } \overline{BC} = (5\hat{i} + 5\hat{j}) - (\hat{i} + 7\hat{k}) = 4\hat{i} + 5\hat{j} - 7\hat{k}$$

$$\Rightarrow BC = \sqrt{90} = a$$

$$\overline{AC} = (5\hat{i} + 5\hat{j}) - (3\hat{i} + 4\hat{j} + 5\hat{k})$$

$$\overline{AC} = (2\hat{i} + \hat{j} - 5\hat{k}) = \sqrt{30} = b$$

In $\triangle ABC$

$$a = \sqrt{90}, b = \sqrt{30}, c = \sqrt{24}$$

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} = \frac{30 + 24 - 90}{2\sqrt{30} \times 24} \\ &= \frac{54 - 90}{2 \times 2\sqrt{15} \times 12} = \frac{-36}{4\sqrt{15} \times 4 \times 3} \\ &= \frac{-9}{\sqrt{5} \times 3 \times 4 \times 3} = \frac{-9}{3 \times 2\sqrt{5}} = -\frac{3}{2\sqrt{5}}\end{aligned}$$

$$\begin{aligned}\cos B &= \frac{c^2 + a^2 - b^2}{2ca} = \frac{24 + 90 - 30}{2 \times \sqrt{24} \times \sqrt{90}} \\ &= \frac{114 - 30}{2 \times \sqrt{24} \times \sqrt{90}} = \frac{84}{2 \times 2\sqrt{6} \times 3\sqrt{10}} \\ &= \frac{7}{\sqrt{6} \times \sqrt{10}} = \frac{7}{2\sqrt{15}}\end{aligned}$$

$$\begin{aligned}\cos C &= \frac{a^2 + b^2 - c^2}{2ab} = \frac{90 + 30 - 24}{2\sqrt{90} \times \sqrt{30}} \\ &= \frac{96}{2 \times 3 \times \sqrt{10} \times \sqrt{3} \times \sqrt{10}} \\ &= \frac{16}{10\sqrt{3}} = \frac{8}{5\sqrt{3}}\end{aligned}$$

$$\text{Circum radius } R = \frac{abc}{4\Delta}$$

Where Δ is the area of triangle which is given by

$$\text{Let } \Delta = \frac{1}{2} |\overline{AB} \times \overline{AC}| \text{ or } \frac{1}{2} |\overline{AB} \times \overline{BC}| \text{ or } \frac{1}{2} |\overline{AC} \times \overline{BC}|$$

$$\Delta = \frac{1}{2} |\overline{AB} \times \overline{AC}| = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -4 & 2 \\ 2 & 1 & -5 \end{vmatrix}$$

$$= \frac{1}{2} |(\hat{i}(20 - 2) - \hat{j}(10 - 4) + \hat{k}(-2 + 8))|$$

$$= \frac{1}{2} |18\hat{i} - 6\hat{j} + 6\hat{k}| = \sqrt{99}$$

Distance between orthocenter and circumcenter is

$$\begin{aligned}R\sqrt{1 - 8 \cos A \cos B \cos C} \\ = \frac{abc}{4\Delta} \sqrt{1 - 8 \left(-\frac{3}{2\sqrt{5}}\right) \left(\frac{7}{2\sqrt{15}}\right) \left(\frac{8}{5\sqrt{3}}\right)}\end{aligned}$$

$$\begin{aligned}
 &= \frac{\sqrt{90}\sqrt{24}\sqrt{30}}{4 \times \sqrt{99}} \sqrt{1 + \frac{7 \times 8 \times 3 \times 8}{2 \times 2 \times 5 \times \sqrt{15}\sqrt{15}}} \\
 &= \frac{30 \times 2 \times 3 \times \sqrt{2}}{4 \times \sqrt{99}} \sqrt{1 + \frac{7 \times 2 \times 3 \times 8}{5 \times 15}} \\
 &= \frac{45\sqrt{2}}{\sqrt{99}} \sqrt{1 + \frac{7 \times 16}{25}} \\
 &= \frac{45\sqrt{2}}{\sqrt{99}} \sqrt{\frac{25 + 112}{25}} \\
 &= \frac{45\sqrt{2}}{\sqrt{99}} \sqrt{\frac{137}{25}} = 3\sqrt{\frac{274}{11}} \\
 &= \frac{9\sqrt{274}}{\sqrt{99}}
 \end{aligned}$$

75. If $\vec{a} \times (\vec{b} \times \vec{c}) + (\vec{a} \cdot \vec{b})\vec{b} = (4 - 2\beta - \sin \alpha)\vec{b} + (\beta^2 - 1)\vec{c}$ and $(\vec{c} \cdot \vec{c})\vec{a} = \vec{c}$, $\vec{b}, \vec{c}$ being non-collinear then

- (1) $\alpha = \frac{\pi}{2}, \beta = -1$ (2) $\alpha = \frac{\pi}{3}, \beta = -1$ (3) $\alpha = \frac{\pi}{3}, \beta = 1$ (4) $\alpha = \frac{\pi}{2}, \beta = 1$

Sol. Answer (4)

$$\vec{a} \times (\vec{b} \times \vec{c}) + (\vec{a} \cdot \vec{b})\vec{b} = (4 - 2\beta - \sin \alpha)\vec{b} + (\beta^2 - 1)\vec{c}$$

$$\Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} + (\vec{a} \cdot \vec{b})\vec{b} = (4 - 2\beta - \sin \alpha)\vec{b} + (\beta^2 - 1)\vec{c}$$

Comparing, we get

$$\vec{a} \cdot \vec{c} + \vec{a} \cdot \vec{b} = 4 - 2\beta - \sin \alpha \quad \dots (i)$$

$$\vec{a} \cdot \vec{b} = 1 - \beta^2 \quad \dots (ii)$$

$$\text{Also } (\vec{c} \cdot \vec{c})\vec{a} = \vec{c}$$

$$\Rightarrow (\vec{c} \cdot \vec{c})(\vec{a} \cdot \vec{c}) = (\vec{c} \cdot \vec{c})$$

$$\Rightarrow ((\vec{a} \cdot \vec{c}) - 1)(\vec{c} \cdot \vec{c}) = 0$$

$$\Rightarrow \vec{a} \cdot \vec{c} = 1, \text{ or } \vec{c} \cdot \vec{c} = 0 \quad \dots (iii)$$

By (i), (ii), (iii)

$$1 + 1 - \beta^2 = 4 - 2\beta - \sin \alpha$$

$$\Rightarrow \beta^2 - 2\beta + 2 - \sin \alpha = 0$$

$$(\beta - 1)^2 + (1 - \sin \alpha) = 0$$

Is valid only when $\beta = 1, \sin \alpha = 1$

$$\Rightarrow \beta = 1, \alpha = \frac{\pi}{2}.$$

[Miscellaneous]

76. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $(\vec{a} \times \vec{b})$ and $\vec{c}$ is 30° , then the value of $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) 2

(4) 3

Sol. Answer (2)

$$|(\vec{a} \times \vec{b}) \times \vec{c}| = |\vec{a} \times \vec{b}| |\vec{c}| \sin 30^\circ$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = \hat{i}(0+2) - \hat{j}(0+2) + \hat{k}(2-1) = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{4+4+1} = 3$$

$$\text{But } |\vec{c} - \vec{a}|^2 = \vec{c}^2 + \vec{a}^2 - 2\vec{a} \cdot \vec{c}$$

$$\Rightarrow 8 = c^2 + 9 - 2c$$

$$\Rightarrow c^2 - 2c + 1 = 0$$

$$\Rightarrow (c-1)^2 = 0$$

$$\Rightarrow c = 1$$

$$\Rightarrow |(\vec{a} \times \vec{b}) \times \vec{c}| = 3 \times 1 \times \sin 30^\circ = \frac{3}{2}$$

77. If $\vec{r} = 3\hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{c} = -2\hat{i} + \hat{j} - 3\hat{k}$ such that $\vec{r} = \lambda\vec{a} + \mu\vec{b} + \gamma\vec{c}$, then

(1) $\mu, \frac{\lambda}{2}, \gamma$ are in A.P.

(2) λ, μ, γ are in A.P.

(3) $\frac{\lambda}{2}, \mu, \frac{\gamma}{2}$ are in A.P.

(4) $\mu, \frac{\lambda}{2}, \gamma$ are in G.P.

Sol. Answer (1)

$$(3\hat{i} + 2\hat{j} - 5\hat{k}) = \lambda(2\hat{i} - \hat{j} + \hat{k}) + \mu(\hat{i} + 3\hat{j} - 2\hat{k}) + \gamma(-2\hat{i} + \hat{j} - 3\hat{k})$$

$$\Rightarrow 2\lambda + \mu - 2\gamma = 3 \quad \dots (i)$$

$$-\lambda + 3\mu + \gamma = 2 \quad \dots (ii)$$

$$\lambda - 2\mu - 3\gamma = -5 \quad \dots (iii)$$

By (i), (ii), and (iii)

$$\mu = 1, \gamma = 2, \lambda = 3$$

$$\mu, \frac{\lambda}{2}, \gamma \text{ are in A.P.}$$

78. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = \vec{0}$, the acute angle between $\vec{a}$ and $\vec{c}$ is

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{4}$ (4) $\frac{2\pi}{3}$

Sol. Answer (1)

$$\vec{a} \times (\vec{a} \times \vec{c}) = -\vec{b}$$

$$(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} = -\vec{b}$$

Let angle between $\vec{a}$ and $\vec{c}$ is θ

$$\Rightarrow (2 \cos \theta)\vec{a} - \vec{c} = -\vec{b}$$

$$\Rightarrow |2 \cos \theta \vec{a} - \vec{c}|^2 = |\vec{b}|^2$$

$$\Rightarrow 4 \cos^2 \theta a^2 + c^2 - 2 \cdot 2 \cos \theta (\vec{a} \cdot \vec{c}) = b^2$$

$$\Rightarrow 4 \cos^2 \theta + 4 - 4 \cos \theta \times 2 \cos \theta = b^2 = 1$$

$$\Rightarrow 4 - 4 \cos^2 \theta = 1$$

$$\Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

79. If the vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other, then a vector $\vec{v}$ in terms of $\vec{a}$ and $\vec{b}$ satisfying the equations $\vec{v} \cdot \vec{a} = 0$, $\vec{v} \cdot \vec{b} = 1$ and $[\vec{v} \vec{a} \vec{b}] = 1$ is

- (1) $\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|^2}$ (2) $\frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|^2}$ (3) $\frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$ (4) $\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$

Sol. Answer (2)

$$\text{Let } \vec{v} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$$

$$\Rightarrow \vec{v} \cdot \vec{a} = x\vec{a} \cdot \vec{a} + y\vec{b} \cdot \vec{a} + z\vec{a} \cdot (\vec{a} \times \vec{b}) \quad \dots (i)$$

$$\Rightarrow \vec{v} \cdot \vec{b} = x\vec{a} \cdot \vec{b} + y\vec{b} \cdot \vec{b} + z\vec{b} \cdot (\vec{a} \times \vec{b}) \quad \dots (ii)$$

$$\Rightarrow \vec{v} \cdot (\vec{a} \times \vec{b}) = x\vec{a} \cdot (\vec{a} \times \vec{b}) + y\vec{b} \cdot (\vec{a} \times \vec{b}) + z(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) \quad \dots (iii)$$

$$\text{but } \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} = \vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{a} \times \vec{b}) = 0$$

also $\vec{v} \cdot \vec{b} = 1$, $\vec{v} \cdot \vec{a} = 0$, $[\vec{v} \vec{a} \vec{b}] = 1$

Putting these values in (i), (ii), (iii) respectively, we get

$$x = 0, y = \frac{1}{b^2}, z = \frac{1}{|\vec{a} \times \vec{b}|^2}$$

$$\Rightarrow \vec{v} = \frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|^2}$$

80. The number of distinct real values of λ , for the vector $-\lambda^2 \hat{i} + \hat{j} + \hat{k}$, $\hat{i} - \lambda^2 \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \lambda^2 \hat{k}$ are coplanar, is [IIT-JEE 2007]

- (1) Zero (2) One (3) Two (4) Three

Sol. Answer (3)

λ is obtained using

$$\begin{vmatrix} -\lambda^2 & 1 & 1 \\ 1 & -\lambda^2 & 1 \\ 1 & 1 & -\lambda^2 \end{vmatrix} = 0$$

81. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct? [IIT-JEE 2007]

- (1) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$ (2) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$
 (3) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$ (4) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

Sol. Answer (2)

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \times (\vec{a} + \vec{b} + \vec{c}) = \vec{a} \times \vec{0} = \vec{0}$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{c} \times \vec{a}$$

$$\therefore \vec{a} \times \vec{b} = \vec{c} \times \vec{a} = \vec{b} \times \vec{c} \neq \vec{0}$$

82. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit vectors $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$. Then, the volume of the parallelopiped is [IIT-JEE 2008]

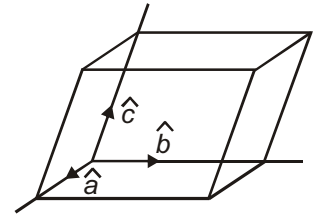
- (1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2\sqrt{2}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{\sqrt{3}}$

Sol. Answer (1)

The volume of the parallelepiped with coterminus edges as $\hat{a}$, $\hat{b}$, $\hat{c}$ is given by $[\hat{a} \hat{b} \hat{c}] = \hat{a} \cdot (\hat{b} \times \hat{c})$

$$\text{Now } [\hat{a} \hat{b} \hat{c}]^2 = \begin{vmatrix} \hat{a} \cdot \hat{a} & \hat{a} \cdot \hat{b} & \hat{a} \cdot \hat{c} \\ \hat{b} \cdot \hat{a} & \hat{b} \cdot \hat{b} & \hat{b} \cdot \hat{c} \\ \hat{c} \cdot \hat{a} & \hat{c} \cdot \hat{b} & \hat{c} \cdot \hat{c} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2}$$

$$\Rightarrow [\hat{a} \hat{b} \hat{c}] = \frac{1}{2}$$



Thus the required volume of the parallelepiped = $\frac{1}{\sqrt{2}}$ cubic units.

83. Let two non-collinear unit vectors $\hat{a}$ and $\hat{b}$ form an acute angle. A point P moves so that at any time t the position vector $\overrightarrow{OP}$ (where O is the origin) is given by $\hat{a} \cos t + \hat{b} \sin t$. When P is farthest from origin O , let M be the length of $\overrightarrow{OP}$ and $\hat{u}$ be the unit vector along $\overrightarrow{OP}$. Then, **[IIT-JEE 2008]**

$$(1) \hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|} \text{ and } M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$$

$$(2) \hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|} \text{ and } M = (1 + \hat{a} \cdot \hat{b})^{\frac{1}{2}}$$

$$(3) \hat{u} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|} \text{ and } M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$$

$$(4) \hat{u} = \frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|} \text{ and } M = (1 + 2\hat{a} \cdot \hat{b})^{\frac{1}{2}}$$

Sol. Answer (1)

$$\overrightarrow{OP} = \hat{a} \cos t + \hat{b} \sin t$$

$$|\overrightarrow{OP}| = \sqrt{(\hat{a} \cdot \hat{a} \cos^2 t + (\hat{b} \cdot \hat{b}) \sin^2 t + 2\hat{a} \cdot \hat{b} \sin t \cos t)}$$

$$|\overrightarrow{OP}| = \sqrt{1 + 2\hat{a} \cdot \hat{b} \sin t \cos t}$$

$$|\overrightarrow{OP}| = \sqrt{1 + \hat{a} \cdot \hat{b} \sin 2t}$$

$$|\overrightarrow{OP}|_{\max} = \sqrt{1 + \hat{a} \cdot \hat{b}} \text{ at } \sin 2t = 1 \Rightarrow t = \frac{\pi}{4}$$

$$\overrightarrow{OP} \left(\text{at } t = \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}} (\hat{a} + \hat{b})$$

$$\text{Unit vector along } \overrightarrow{OP} \text{ at } \left(t = \frac{\pi}{4} \right) = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|}$$

84. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$, then [IIT-JEE 2009]

- (1) $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar (2) $\vec{b}, \vec{c}, \vec{d}$ are non-coplanar
 (3) $\vec{b}, \vec{d}$ are non-parallel (4) $\vec{a}, \vec{d}$ are parallel and $\vec{b}, \vec{c}$ are parallel

Sol. Answer (3)

Let the angle between vectors $\hat{a}$ and $\hat{b}$, $\hat{b}$ and $\hat{c}$, $\hat{c}$ and $\hat{d}$ be α , β and γ respectively

$$\Rightarrow (\hat{a} \times \hat{b}) \cdot (\hat{c} \times \hat{d}) = 1$$

$$(\sin \alpha \hat{n}_1) \cdot (\sin \beta \hat{n}_2) = 1$$

$$\Rightarrow \sin \alpha \sin \beta (\hat{n}_1 \cdot \hat{n}_2) = 1$$

Let the angle between $\hat{n}_1$ & $\hat{n}_2$ be θ

$$\Rightarrow \sin \alpha \sin \beta \cos \theta = 1$$

This is possible when

$$\sin \alpha = \sin \beta = \cos \theta = 1$$

...(i)

$$\text{Also, } \vec{a} \cdot \vec{c} = \frac{1}{2}$$

$$\Rightarrow 1 \times 1 \times \cos \gamma = \frac{1}{2}$$

$$\Rightarrow \gamma = \frac{\pi}{3}$$

...(ii)

From (i), we have

$$\alpha = \beta = \frac{\pi}{2}$$

$$\theta = 0^\circ$$

As $\theta = 90^\circ$, then $\hat{a}, \hat{b}, \hat{c}, \hat{d}$ will be coplanar. Option (1) and (2) are wrong. Option (4) also is not true. Hence option (3) is correct.

85. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}$, $3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral $PQRS$ must be a [IIT-JEE 2010]

- (1) Parallelogram, which is neither a rhombus nor a rectangle
 (2) Square
 (3) Rectangle, but not a square
 (4) Rhombus, but not a square

Sol. Answer (1)

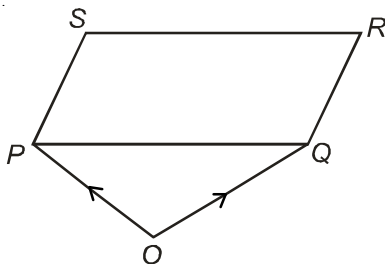
Let O be the origin

then $\overrightarrow{OP} = -2\hat{i} - \hat{j}$

$\overrightarrow{OQ} = 4\hat{i}$

$\overrightarrow{OR} = 3\hat{i} + 3\hat{j}$

$\overrightarrow{OS} = -3\hat{i} + 2\hat{j}$



Here we have

$\overrightarrow{PQ} = 6\hat{i} + \hat{j}$

$\overrightarrow{QR} = -\hat{i} + 3\hat{j}$

$\overrightarrow{RS} = -6\hat{i} - \hat{j}$

and $\overrightarrow{PS} = -\hat{i} + 3\hat{j}$

$\overrightarrow{PR} = 5\hat{i} + 4\hat{j}$

$\overrightarrow{QS} = -7\hat{i} + 2\hat{j}$

$\overrightarrow{PR} \cdot \overrightarrow{QS} = -35 + 8 = -27 \neq 0$

Diagonals are not perpendicular

and $|\overrightarrow{PQ}| = |\overrightarrow{RS}|$, $|\overrightarrow{QR}| = |\overrightarrow{PS}|$

Hence $PQRS$ is a parallelogram which is neither a rhombus nor a rectangle.

86. Two adjacent sides of a parallelogram $ABCD$ are given by $\overrightarrow{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\overrightarrow{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD' . If AD' makes a right angle with the side AB , then the cosine of the angle α is given by **[IIT-JEE 2010]**

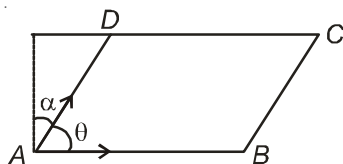
(1) $\frac{8}{9}$

(2) $\frac{\sqrt{17}}{9}$

(3) $\frac{1}{9}$

(4) $\frac{4\sqrt{5}}{9}$

Sol. Answer (2)



$\overrightarrow{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$

$$\overrightarrow{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$$

Angle ' θ ' between $\overrightarrow{AB}$ and $\overrightarrow{AD}$

$$= \cos(\theta) = \frac{|\overrightarrow{AB} \cdot \overrightarrow{AD}|}{|\overrightarrow{AB}| |\overrightarrow{AD}|} = \frac{|-2 + 20 + 22|}{(15)(3)} = \frac{8}{9}$$

$$\Rightarrow \sin(\theta) = \frac{\sqrt{17}}{9}$$

$$\therefore \alpha + \theta = 90$$

$$\cos(\alpha) = \cos(90 - \theta) = \sin(\theta) = \frac{\sqrt{17}}{9}$$

87. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on $\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by [IIT-JEE 2011]

(1) $\hat{i} - 3\hat{j} + 3\hat{k}$ (2) $-3\hat{i} - 3\hat{j} - \hat{k}$ (3) $3\hat{i} - \hat{j} + 3\hat{k}$ (4) $\hat{i} + 3\hat{j} - 3\hat{k}$

Sol. Answer (3)

A vector in the plane of $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ is given by

$$\begin{aligned} & \vec{a} + \lambda \vec{b} \\ &= (\hat{i} + \hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}) \\ &= (1 + \lambda)\hat{i} + (1 - \lambda)\hat{j} + (1 + \lambda)\hat{k} \end{aligned}$$

whose projection on $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ is given to be $\frac{1}{\sqrt{3}}$, hence

$$\frac{(1 + \lambda) - (1 - \lambda) - (1 + \lambda)}{\sqrt{3}} = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow \lambda - 1 = \pm 1$$

$$\Rightarrow \lambda = 0 \text{ or } \lambda = 2$$

But $\lambda = 0$ does not serve our purpose

$$\text{when } \lambda = 2, \vec{v} = 3\hat{i} - \hat{j} + 3\hat{k}$$

88. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a} + \vec{b}| = \sqrt{29}$ and $\vec{a} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{b}$, then a possible value of $(\vec{a} + \vec{b}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$ is [IIT-JEE 2012]

(1) 0 (2) 3
(3) 4 (4) 8

Sol. Answer (3)

$$\vec{a} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) + \vec{b} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = \vec{0}$$

$$\Rightarrow (\vec{a} + \vec{b}) \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = \vec{0}$$

$$\Rightarrow \vec{a} + \vec{b} = \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$\text{Given } |\vec{a} + \vec{b}| = \sqrt{29}$$

$$\Rightarrow \lambda = 1$$

$$\text{So, } \vec{a} + \vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\text{Now, } (\vec{a} + \vec{b}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k}) = 4$$

89. Let $\vec{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\vec{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\vec{PT}$, $\vec{PQ}$ and $\vec{PS}$ is

[JEE(Advanced) 2013]

(1) 5

(2) 20

(3) 10

(4) 30

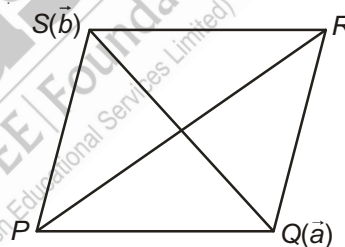
Sol. Answer (3)

$$\vec{PR} = \vec{a} + \vec{b} = 3\hat{i} + \hat{j} - 2\hat{k} = \vec{d}_1$$

$$\vec{SQ} = \vec{a} - \vec{b} = \hat{i} - 3\hat{j} - 4\hat{k} = \vec{d}_2$$

$$\Rightarrow \vec{a} = 2\hat{i} - \hat{j} - 3\hat{k}$$

$$\Rightarrow \vec{b} = \hat{i} + 2\hat{j} + \hat{k}$$



$$\therefore \text{Volume of the required parallelepiped} = \begin{vmatrix} 2 & -1 & -3 \\ 1 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= |2(6 - 2) + 1(3 - 1) - 3(2 - 2)|$$

$$= 10 \text{ cubic units.}$$

OR

$$\text{Area of parallelogram} = \frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$$

$$\therefore \text{Volume of parallelepiped} = \frac{1}{2} \begin{vmatrix} 3 & 1 & -2 \\ 1 & -3 & -4 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= 10 \text{ cubic units.}$$

90. Let O be the origin and let PQR be an arbitrary triangle. The point S is such that

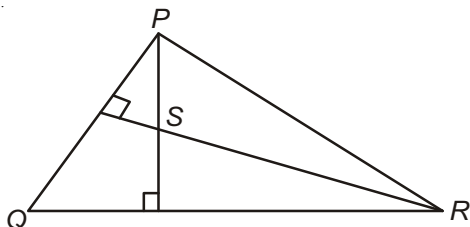
$$\overrightarrow{OP} \cdot \overrightarrow{OQ} + \overrightarrow{OR} \cdot \overrightarrow{OS} = \overrightarrow{OR} \cdot \overrightarrow{OP} + \overrightarrow{OQ} \cdot \overrightarrow{OS} = \overrightarrow{OQ} \cdot \overrightarrow{OR} + \overrightarrow{OP} \cdot \overrightarrow{OS}$$

Then the triangle PQR has S as its

[JEE(Advanced) 2017]

- (1) Incentre (2) Circumcentre (3) Orthocenter (4) Centroid

Sol. Answer (3)



$$\overrightarrow{OP} \cdot \overrightarrow{OQ} + \overrightarrow{OR} \cdot \overrightarrow{OS} = \overrightarrow{OR} \cdot \overrightarrow{OP} + \overrightarrow{OQ} \cdot \overrightarrow{OS} = \overrightarrow{OQ} \cdot \overrightarrow{OR} + \overrightarrow{OP} \cdot \overrightarrow{OS}$$

$$\overrightarrow{OP} \cdot \overrightarrow{OQ} + \overrightarrow{OR} \cdot \overrightarrow{OS} = \overrightarrow{OR} \cdot \overrightarrow{OP} + \overrightarrow{OQ} \cdot \overrightarrow{OS}$$

$$\Rightarrow \overrightarrow{OP} \cdot (\overrightarrow{OQ} - \overrightarrow{OR}) + \overrightarrow{OS} \cdot (\overrightarrow{OR} - \overrightarrow{OQ}) = 0$$

$$\Rightarrow \overrightarrow{RQ} \cdot (\overrightarrow{OP} - \overrightarrow{OS}) = 0$$

$$\Rightarrow \overrightarrow{RQ} \cdot \overrightarrow{SP} = 0$$

$$\Rightarrow \overrightarrow{RQ} \perp \overrightarrow{SP}$$

$$\text{and similarly from } \overrightarrow{OR} \cdot \overrightarrow{OP} + \overrightarrow{OQ} \cdot \overrightarrow{OS} = \overrightarrow{OQ} \cdot \overrightarrow{OR} + \overrightarrow{OP} \cdot \overrightarrow{OS}$$

$$\overrightarrow{SR} \perp \overrightarrow{PQ}$$

$\therefore S$ is the orthocentre.

SECTION - B

Objective Type Questions (More than one options are correct)

1. Let $\vec{\alpha} = a\hat{i} + b\hat{j} + c\hat{k}$, $\vec{\beta} = b\hat{i} + c\hat{j} + a\hat{k}$ and $\vec{\gamma} = c\hat{i} + a\hat{j} + b\hat{k}$ be three coplanar vectors with $a \neq b \neq c$ and

$\vec{V} = \hat{i} + \hat{j} + \hat{k}$. Then $\vec{V}$ is perpendicular to

- (1) $\vec{\alpha} + \vec{\beta}$ (2) $\vec{\beta} + \vec{\gamma}$ (3) $\vec{\gamma} + \vec{\alpha}$ (4) $\vec{\alpha} + \vec{\beta} + \vec{\gamma}$

Sol. Answer (1, 2, 3, 4)

For coplanar vectors

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$\Rightarrow -(a^3 + b^3 + c^3 - 3abc) = 0$$

$$\Rightarrow (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0$$

$$\Rightarrow a+b+c = 0$$

$$\vec{\alpha} + \vec{\beta} = (a+b)\hat{i} + (b+c)\hat{j} + (c+a)\hat{k}$$

$$(\vec{\alpha} + \vec{\beta}) \cdot (\hat{i} + \hat{j} + \hat{k}) = 2(\vec{a} + \vec{b} + \vec{c}) = 0$$

$$\text{Hence } \vec{\alpha} + \vec{\beta} \perp \vec{v}$$

$$\text{Similarly } (\vec{\beta} + \vec{\gamma}) \cdot \vec{v} = 0$$

$$\text{and } (\vec{\gamma} + \vec{\alpha}) \cdot \vec{v} = 0 \text{ so } (\vec{\alpha} + \vec{\beta} + \vec{\gamma}) \cdot (\hat{i} + \hat{j} + \hat{k}) = 0$$

Hence options (1), (2), (3), (4) are correct.

2. If $\vec{a} + 2\vec{b} + 3\vec{c} = 0$, then $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ is equal to

(1) $6(\vec{b} \times \vec{c})$ (2) $2(\vec{a} \times \vec{b})$ (3) $3(\vec{c} \times \vec{a})$ (4) 0

Sol. Answer (1, 2, 3)

$$\text{Let } y = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$$

(1) if $\vec{a} = -(2\vec{b} + 3\vec{c})$

$$\begin{aligned} y &= \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} \\ &= -(2\vec{b} + 3\vec{c}) \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times (-2\vec{b} - 3\vec{c}) \\ &= 3\vec{b} \times \vec{c} + \vec{b} \times \vec{c} + 2\vec{b} \times \vec{c} = 6(\vec{b} \times \vec{c}) \end{aligned}$$

(2) If $\vec{c} = -\frac{\vec{a} + 2\vec{b}}{3}$

$$\begin{aligned} \Rightarrow y &= \vec{a} \times \vec{b} + \vec{b} \times \left(\frac{-\vec{a} - 2\vec{b}}{3} \right) + \left(\frac{-\vec{a} - 2\vec{b}}{3} \right) \times \vec{a} \\ \Rightarrow y &= 2(\vec{a} \times \vec{b}) \end{aligned}$$

(3) If $\vec{b} = \frac{-(\vec{a} + 3\vec{c})}{2} \Rightarrow y = \vec{a} \times \left(\frac{-\vec{a} - 3\vec{c}}{2} \right) + \left(\frac{-\vec{a} - 3\vec{c}}{2} \right) \times \vec{c} + (\vec{c} \times \vec{a})$

Hence options (1), (2), (3) are correct.

3. The vector $(\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b}$ is perpendicular to

(1) $\vec{b}$ (2) $\vec{a}$ (3) $\vec{c}$ (4) $\vec{b} \times \vec{c}$

Sol. Answer (2, 4)

$$\text{Let } \vec{d} = (\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b}$$

$$\vec{d} \cdot \vec{a} = (\vec{a} \cdot \vec{b})(\vec{c} \cdot \vec{a}) - (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{a}) = 0$$

Hence $\vec{a}$ is perpendicular to $\vec{d}$

$$\begin{aligned}
 \vec{d} \cdot (\vec{b} \times \vec{c}) &= ((\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} - \vec{c})\vec{b}) \cdot (\vec{b} \times \vec{c}) \\
 &= (\vec{a} \cdot \vec{b})[\vec{c} \cdot (\vec{b} \times \vec{c})] - (\vec{a} \cdot \vec{c})[\vec{b} \cdot (\vec{b} \times \vec{c})] \\
 &= (\vec{a} \cdot \vec{b})(0) - (\vec{a} \cdot \vec{c})(0) = 0
 \end{aligned}$$

Hence vector $\vec{d}$ is perpendicular to $\vec{a}$ and $\vec{b} \times \vec{c}$.

Hence options (2, 4) are correct.

4. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors such that $[\vec{a} \vec{b} \vec{c}] = |\vec{a}| |\vec{b}| |\vec{c}|$ then

$$(1) \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c}$$

$$(2) \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$$

$$(3) \vec{c} \cdot \vec{a} = \vec{a} \cdot \vec{b}$$

$$(4) \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$$

Sol. Answer (1, 2, 3, 4)

$$[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{a} \cdot (|\vec{b}| |\vec{c}| \sin \theta) \hat{n}$$

Where ' θ ' is angle between $\vec{b}$ and $\vec{c}$ and $\hat{n}$ is the unit vector perpendicular to the plane containing $\vec{b}$ and $\vec{c}$. If the angle between $\vec{a}$ and $\hat{n}$ is ϕ .

$$\begin{aligned}
 \Rightarrow [\vec{a} \vec{b} \vec{c}] &= |\vec{a}| |\vec{b}| |\vec{c}| |\hat{n}| \sin \theta \cos \phi \\
 &= |\vec{a}| |\vec{b}| |\vec{c}| \sin \theta \cos \phi
 \end{aligned}$$

$$\text{But } [\vec{a} \vec{b} \vec{c}] = |\vec{a}| |\vec{b}| |\vec{c}|$$

$$\Rightarrow \sin \theta \cos \phi = 1 \quad \dots (i)$$

Equation (i) is valid only when $\theta = 90^\circ$, $\phi = 0^\circ$

Hence vectors $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular

$$\Rightarrow \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$$

$$\text{and } \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$$

Hence options (1), (2), (3) & (4) are correct

5. Which of the following expressions are meaningful?

$$(1) \vec{u} \times (\vec{v} \times \vec{w})$$

$$(2) (\vec{u} \cdot \vec{v}) \cdot \vec{w}$$

$$(3) (\vec{u} \cdot \vec{v}) \vec{w}$$

$$(4) \vec{u} \times (\vec{v} \cdot \vec{w})$$

Sol. Answer (1, 3)

Only $\vec{u} \times (\vec{v} \times \vec{w})$ and $(\vec{u} \cdot \vec{v}) \vec{w}$ are meaningful because dot product and cross product is valid only in vectors.

6. If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -2\hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 10\hat{j} - \hat{k}$ and $\vec{a} \times (\vec{b} \times \vec{c}) = u\vec{a} + v\vec{b} + w\vec{c}$, then

$$(1) w = 0$$

$$(2) w = -3$$

$$(3) u = 0$$

$$(4) v = 17$$

Sol. Answer (2, 3, 4)

7. The vector(s) which is/are coplanar with vectors $\hat{i} + \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} + \hat{k}$, and perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ is/are [IIT-JEE 2011]

(1) $\hat{j} - \hat{k}$ (2) $-\hat{j} + \hat{k}$ (3) $\hat{i} - \hat{j}$ (4) $-\hat{j} + \hat{k}$

Sol. Answer (1, 4)

Required vector will be parallel or antiparallel to the vectors

$$\begin{aligned} & (\hat{i} + \hat{j} + \hat{k}) \times ((\hat{i} + \hat{j} + 2\hat{k}) \times (\hat{i} + 2\hat{j} + \hat{k})) \\ &= 4(\hat{i} + \hat{j} + 2\hat{k}) - 4(\hat{i} + 2\hat{j} + \hat{k}) \\ &= -4\hat{j} + 4\hat{k} \end{aligned}$$

8. Let $\vec{x}$, $\vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y} \times \vec{z}$ and $\vec{b}$ is a nonzero vector perpendicular to $\vec{y}$ and $\vec{z} \times \vec{x}$, then

[JEE(Advanced)-2014]

(1) $\vec{b} = (\vec{b} \cdot \vec{z})(\vec{z} - \vec{x})$ (2) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{y} - \vec{z})$ (3) $\vec{a} \cdot \vec{b} = -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z})$ (4) $\vec{a} = (\vec{a} \cdot \vec{y})(\vec{z} - \vec{y})$

Sol. Answer (1, 2, 3)

Given that $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{z} = \vec{z} \cdot \vec{x} = 1$ and $\vec{a} = \lambda(\vec{x} \times (\vec{y} \times \vec{z}))$, $\vec{b} = \mu(\vec{y} \times (\vec{z} \times \vec{x}))$

$$= \lambda(\vec{y} - \vec{z}) \quad = \mu(\vec{z} - \vec{x})$$

$$\therefore \vec{a} \cdot \vec{y} = \lambda(2 - 1) = \lambda, \vec{b} \cdot \vec{z} = \mu$$

$$(\vec{a} \cdot \vec{y})(\vec{y} - \vec{z}) = \lambda(\vec{y} - \vec{z}) = \vec{a} \quad \text{(Option 2 is correct)}$$

$$(\vec{b} \cdot \vec{z})(\vec{z} - \vec{x}) = \mu(\vec{z} - \vec{x}) = \vec{b} \quad \text{(Option 1 is correct)}$$

$$(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z}) = \lambda\mu$$

$$\text{and } \vec{a} \cdot \vec{b} = \lambda\mu(\vec{y} - \vec{z}) \cdot (\vec{z} - \vec{x})$$

$$= \lambda\mu(1 - 1 - 2 + 1)$$

$$= -\lambda\mu$$

$$= -(\vec{a} \cdot \vec{y})(\vec{b} \cdot \vec{z}) \quad \text{(Option 3 is correct)}$$

$$(\vec{a} \cdot \vec{y})(\vec{z} - \vec{y}) = \lambda(\vec{z} - \vec{y}) \neq \vec{a}$$

$\therefore$ (1, 2, 3) are correct.

9. Let ΔPQR be a triangle. Let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 24$, then which of the following is (are) true? [JEE(Advanced)-2015]

(1) $\frac{|\vec{c}|^2}{2} - |\vec{a}| = 12$ (2) $\frac{|\vec{c}|^2}{2} + |\vec{a}| = 30$ (3) $|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 48\sqrt{3}$ (4) $\vec{a} \cdot \vec{b} = -72$

Sol. Answer (1, 3, 4)

$$|\vec{a}| = 12, |\vec{b}| = 4\sqrt{3}, \vec{b} \cdot \vec{c} = 24$$

$$\vec{a} + \vec{b} + \vec{c} = 0 \quad \text{(Sides of triangle)}$$

$$\vec{b} + \vec{c} = -\vec{a}$$

$$(\vec{b} + \vec{c}) \cdot (\vec{b} + \vec{c}) = -\vec{a} \cdot -\vec{a}$$

$$|\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b} \cdot \vec{c} = |\vec{a}|^2$$

$$48 + |\vec{c}|^2 + 2 \times 24 = 12^2$$

$$|\vec{c}|^2 = 48$$

$$\frac{|\vec{c}|^2}{2} - |\vec{a}| = \frac{48}{2} - 12 = 12$$

$$\frac{|\vec{c}|^2}{2} + |\vec{a}| = \frac{48}{2} + 12 = 36$$

$$|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = |\vec{a} \times \vec{b} - \vec{a} \times \vec{c}| = |\vec{a} \times (\vec{b} - \vec{c})|$$

$$\vec{a} = -\vec{b} - \vec{c}$$

$$= |(-\vec{b} - \vec{c}) \times (\vec{b} - \vec{c})| = |(\vec{b} - \vec{c}) \times (\vec{b} + \vec{c})|$$

$$= 2|\vec{b} \times \vec{c}|$$

$$= 2\sqrt{|\vec{b}|^2 |\vec{c}|^2 - (\vec{b} \cdot \vec{c})^2} = 2\sqrt{48 \times 48 - (24)^2}$$

$$= 2 \times 24\sqrt{4-1} = 48\sqrt{3}$$

$$\vec{a} + \vec{b} = -\vec{c}$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = -\vec{c} \cdot -\vec{c}$$

$$|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{c}|^2$$

$$\vec{a} \cdot \vec{b} = \frac{-|\vec{a}|^2}{2} = \frac{-12^2}{2} = -72$$

10. Consider a pyramid $OPQRS$ located in the first octant ($x \geq 0, y \geq 0, z \geq 0$) with O as origin, and OP and OR along the x -axis and the y -axis, respectively. The base $OPQR$ of the pyramid is a square with $OP = 3$. The point S is directly above the mid-point T of diagonal OQ such that $TS = 3$. Then **[JEE(Advanced)-2016]**

(1) The acute angle between OQ and OS is $\frac{\pi}{3}$

(2) The equation of the plane containing the triangle OQS is $x - y = 0$

(3) The length of the perpendicular from P to the plane containing the triangle OQS is $\frac{3}{\sqrt{2}}$

(4) The perpendicular distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

Sol. Answer (2, 3, 4)

(1) DR of OQ(1, 1, 0)

DR of OS(1, 1, 2)

So, $\cos\theta = \frac{1+1}{\sqrt{2}\cdot\sqrt{6}} = \frac{1}{\sqrt{3}}$

(2) $\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix} = 2\hat{i} - 2\hat{j}$

So, equation of plane is $x - y = 0$

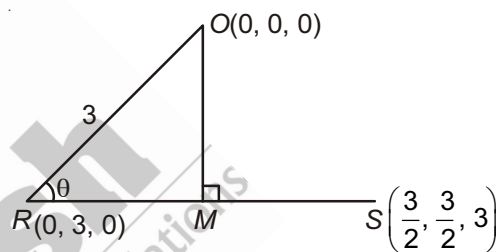
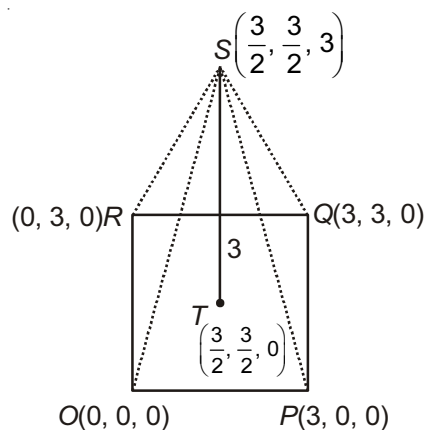
(3) Perpendicular distance = $\frac{3-0}{\sqrt{2}} = \frac{3}{\sqrt{2}}$

(4) DR of OR(0, 1, 0)

DR of RS(1, -1, 2)

So, $\cos\theta = \frac{1}{\sqrt{6}}$

$\Rightarrow OM = OR\sin\theta = 3 \cdot \frac{\sqrt{5}}{\sqrt{6}} = \sqrt{\frac{15}{2}}$



11. Let $\hat{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ be a unit vector in $\mathbb{R}^3$ and $\hat{w} = \frac{1}{\sqrt{6}}(\hat{i} + \hat{j} + 2\hat{k})$. Given that there exists a vector $\vec{v}$ in $\mathbb{R}^3$ such that $|\hat{u} \times \vec{v}| = 1$ and $\hat{w} \cdot (\hat{u} \times \vec{v}) = 1$. Which of the following statement(s) is(are) correct?

[JEE(Advanced)-2016]

- (1) There is exactly one choice for such $\vec{v}$ (2) There are infinitely many choices for such $\vec{v}$
 (3) If $\hat{u}$ lies in the xy-plane then $|u_1| = |u_2|$ (4) If $\hat{u}$ lies in the xz-plane then $2|u_1| = |u_3|$

Sol. Answer (2, 3)

$\hat{w} \cdot (\hat{u} \times \vec{v}) = 1 \Rightarrow \hat{u} \times \vec{v} = \hat{w}$

Let $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$ then

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ x & y & z \end{vmatrix} = \frac{1}{\sqrt{6}}(\hat{i} + \hat{j} + 2\hat{k})$$

$$\Rightarrow zu_2 - yu_3 = \frac{1}{\sqrt{6}}$$

$$\text{and } -zu_1 + xu_3 = \frac{1}{\sqrt{6}}$$

$$\text{and } yu_1 - xu_2 = \frac{2}{\sqrt{6}}$$

$$D = \begin{vmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} \frac{1}{\sqrt{6}} & -u_3 & u_2 \\ \frac{1}{\sqrt{6}} & 0 & -u_1 \\ \frac{2}{\sqrt{6}} & u_1 & 0 \end{vmatrix} = \frac{u_1}{\sqrt{6}}(u_1 + u_2 + 2u_3)$$

Similarly,

$$D_2 = \frac{u_2}{\sqrt{6}}(u_1 + u_2 + 2u_3)$$

$$\text{and } D_3 = \frac{u_3}{\sqrt{6}}(u_1 + u_2 + 2u_3)$$

As it is given that there exist a vector $\vec{v}$

So the equations can have infinite solution only.

$$\Rightarrow u_1 + u_2 + 2u_3 = 0$$

$$\text{If } u_3 = 0 \Rightarrow u_1 + u_2 = 0 \Rightarrow |u_1| = |u_2|$$

$$\text{If } u_2 = 0 \Rightarrow u_1 + 2u_3 = 0 \Rightarrow |u_1| = 2|u_3|$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Let k be the length of any edge of a regular tetrahedron (all edges are equal in length). The angle between a line and a plane is equal to the complement of the angle between the line and the normal to the plane whereas the angle between two plane is equal to the angle between the normals. Let O be the origin and A, B and C vertices with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively of the regular tetrahedron.

1. The angle between any edge and a face not containing the edge is

$$(1) \cos^{-1}\left(\frac{1}{2}\right) \quad (2) \cos^{-1}\left(\frac{1}{4}\right) \quad (3) \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \quad (4) \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

2. The angle between any two faces is

$$(1) \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \quad (2) \cos^{-1}\left(\frac{1}{4}\right) \quad (3) \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \quad (4) \cos^{-1}\left(\frac{1}{3}\right)$$

3. The value of $[\vec{a} \ \vec{b} \ \vec{c}]^2$ is

- (1) k^6 (2) $\frac{1}{2}k^6$ (3) $\frac{1}{3}k^6$ (4) $\frac{1}{4}k^6$

Solution of Comprehension-I

1. Answer (3)

2. Answer (4)

3. Answer (2)

$$|\vec{a}| = |\vec{b}| = |\vec{c}| = k \text{ (Let)}$$

If we take base ABC then perpendicular vector to ABC is $\pm(\vec{b} - \vec{a}) \times (\vec{c} - \vec{b})$

$$\vec{n} = \pm(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$$

Let we take edge $\vec{OA}$ then angle between $\vec{n}$ and $\vec{OA}$ is the angle between $\vec{n}$ and $\vec{a}$

$$\theta = \sin^{-1} \left(\frac{\vec{n} \cdot \vec{a}}{|\vec{n}| |\vec{a}|} \right) = \sin^{-1} \left(\frac{[\vec{a} \ \vec{b} \ \vec{c}]}{|\vec{n}| |\vec{a}|} \right)$$

As we know that

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} a.a & a.b & a.c \\ b.a & b.b & b.c \\ c.a & c.b & c.c \end{vmatrix} = \begin{vmatrix} k^2 & k^2 \cos 60^\circ & k^2 \cos 60^\circ \\ k^2 \cos 60^\circ & k^2 & k^2 \cos 60^\circ \\ k^2 \cos 60^\circ & k^2 \cos 60^\circ & k^2 \end{vmatrix}$$

$$= k^6 \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = K^6 \left[1 \left(1 - \frac{1}{4} \right) - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{4} \right) + \frac{1}{2} \left(\frac{1}{4} - \frac{1}{2} \right) \right]$$

$$= K^6 \left[\frac{3}{4} - \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \left(-\frac{1}{4} \right) \right]$$

$$= K^6 \left[\frac{3}{4} - \frac{1}{4} \right] = \frac{k^6}{2}$$

$$\Rightarrow [abc] = \frac{k^3}{\sqrt{2}}$$

$\Rightarrow$ As a regular tetrahedron the angle between two faces is

$$= \sin^{-1} \frac{|(\vec{a} \times \vec{b}) \times \vec{b} \times \vec{c}|}{|\vec{a} \times \vec{b}| |\vec{b} \times \vec{c}|} = \sin^{-1} \frac{[bca] |b|}{|\vec{a} \times \vec{b}| |\vec{b} \times \vec{c}|}$$

$$= \sin^{-1} \frac{2\sqrt{2}}{3} = \cos^{-1} \frac{1}{3}$$

$$\vec{n} = k^2 \times \frac{\sqrt{3}}{2} [\vec{n}_1 + \vec{n}_2 + \vec{n}_3]$$

$$|\vec{n}| = k^2 \times \frac{\sqrt{3}}{2} \sqrt{1+1+1-2\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{2}\right)} = k^2 \times \frac{\sqrt{3}}{2}$$

$$\theta = \sin^{-1} \left(\frac{\frac{k^3}{\sqrt{2}}}{k^2 \frac{\sqrt{3}}{2} \cdot k} \right) = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

Comprehension-II

If $\vec{b} \neq 0$, then every vector $\vec{a}$ can be written in a unique manner as the sum of a vector $\vec{a}_p$ parallel to $\vec{b}$ and a vector $\vec{a}_q$ perpendicular to $\vec{b}$. If $\vec{a}$ is parallel to $\vec{b}$ then $\vec{a}_q = 0$ and $\vec{a}_p = \vec{a}$. The vector $\vec{a}_p$ is called the projection of $\vec{a}$ on $\vec{b}$ and is denoted by $\text{proj } \vec{b}(\vec{a})$. Since $\text{proj } \vec{b}(\vec{a})$ is parallel to $\vec{b}$, it is a scalar multiple of the unit vector in the direction of $\vec{b}$ i.e.,

$$\text{proj } \vec{b}(\vec{a}) = \lambda \vec{U}_b \quad \left(\vec{U}_b = \frac{\vec{b}}{|\vec{b}|} \right)$$

The scalar λ is called the component of $\vec{a}$ in the direction of $\vec{b}$ and is denoted by $\text{comp } \vec{b}(\vec{a})$. In fact

$$\text{proj } \vec{b}(\vec{a}) = (\vec{a} \cdot \vec{U}_b) \vec{U}_b \text{ and } \text{comp } \vec{b}(\vec{a}) = \vec{a} \cdot \vec{U}_b$$

1. If $\vec{a} = -2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 4\hat{i} - 3\hat{j} + \hat{k}$ then $\text{proj } \vec{b}(\vec{a})$ is

(1) $4\hat{i} - 3\hat{j} + 2\hat{k}$ (2) $-\frac{5}{13}(4\hat{i} - 3\hat{j} + \hat{k})$ (3) $\frac{5}{13}(4\hat{i} - 3\hat{j} + \hat{k})$ (4) $-\frac{4}{11}(4\hat{i} - 3\hat{j} + 2\hat{k})$

Sol. Answer (2)

$$\begin{aligned} \text{Projection } \vec{b}(\vec{a}) &= \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \hat{b} \\ &= \frac{-8-3+1}{\sqrt{26}} \left(\frac{4\hat{i}-3\hat{j}+\hat{k}}{\sqrt{26}} \right) = -\frac{5}{13}(4\hat{i}-3\hat{j}+\hat{k}) \end{aligned}$$

2. If $\vec{a} = -2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 4\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{a} = \vec{a}_p + \vec{a}_q$ then $\vec{a}_q$ is equal to

(1) $\frac{1}{13}(-3\hat{i} + \hat{j} + 9\hat{k})$ (2) $\frac{1}{13}(-3\hat{i} - \hat{j} + 4\hat{k})$ (3) $\frac{2}{13}(-3\hat{i} - \hat{j} + 9\hat{k})$ (4) $\frac{2}{13}(3\hat{i} + \hat{j} - 9\hat{k})$

Sol. Answer (3)

$$\begin{aligned}\vec{a}_p &= \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \right) \hat{b} = \frac{(-8-3+1)}{\sqrt{26}} \cdot \frac{(4\hat{i}-3\hat{j}+\hat{k})}{\sqrt{26}} \\ &= -\frac{5}{13}(4\hat{i}-3\hat{j}+\hat{k})\end{aligned}$$

$$\begin{aligned}\vec{a}_q &= \vec{a} - \vec{a}_p = (-2\hat{i} + \hat{j} + \hat{k}) + \frac{5}{13}(4\hat{i} - 3\hat{j} + \hat{k}) \\ &= \frac{-6\hat{i} - 2\hat{j} + 18\hat{k}}{13} \\ &= \frac{2}{13}(-3\hat{i} - \hat{j} + 9\hat{k})\end{aligned}$$

3. If $\vec{a} = -2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} - 3\hat{j} + \hat{k}$ then $\text{comp } \vec{b}(\vec{a})$ is

(1) $-\frac{5\sqrt{26}}{13}$ (2) $\frac{5\sqrt{26}}{13}$ (3) $5\sqrt{26}$ (4) $\left(-\frac{5}{11}\right)\sqrt{17}$

Sol. Answer (1)

$$\text{Comp } \vec{b}(\vec{a}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{-8-3+1}{\sqrt{26}} = \frac{-5\sqrt{26}}{13}$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : Let $\vec{a}, \vec{b}$ be two vectors such that $\vec{a} \cdot \vec{b} = 0$, then $\vec{a}$ and $\vec{b}$ are perpendicular.

and

STATEMENT-2 : Two non-zero vectors are perpendicular if and only if their dot product is zero.

Sol. Answer (1)

2. STATEMENT-1 : Let $P(\vec{a})$, $Q(\vec{b})$ and $R(\vec{c})$ be three points such that $2\vec{a} + 3\vec{b} - 5\vec{c} = 0$. Then the vector area of the ΔPQR is a null vector.

and

STATEMENT-2 : Three collinear points form a triangle with zero area.

Sol. Answer (1)

$2\vec{a} + 3\vec{b} - 5\vec{c} = 0$ and $2 + 3 - 5 = 0$ hence points are collinear and the area of the triangle is zero.

3. STATEMENT-1 : The vector $\hat{i}$ bisects the angle between the vectors $\hat{i} - 2\hat{j} - 2\hat{k}$ and $\hat{i} + 2\hat{j} + 2\hat{k}$.

and

STATEMENT-2 : The vector along the angle bisector of the vector $\vec{a}$ and $\vec{b}$ is given by $\pm \left(\frac{\vec{a}}{|\vec{a}|} \pm \frac{\vec{b}}{|\vec{b}|} \right)$ where $|\vec{a}|, |\vec{b}| \neq 0$.

Sol. Answer (3)

4. Let $\vec{u}$ and $\vec{v}$ be unit vectors inclined at an angle θ such that for some vector $\vec{w}$, $\vec{w} + \vec{w} \times \vec{u} = \vec{v}$.

STATEMENT-1 : $\vec{u} \cdot \vec{w} = \cos \theta$.

and

STATEMENT-2 : $|\vec{u} \times \vec{v}| = \sin \theta$.

Sol. Answer (2)

$$\vec{w} + \vec{w} \times \vec{u} = \vec{v}$$

dot product by $\vec{u}$

$$\vec{u} \cdot \vec{w} + \vec{u} \cdot (\vec{w} \times \vec{u}) = \vec{u} \cdot \vec{v}$$

$$\Rightarrow \vec{u} \cdot \vec{w} = |\vec{u}| |\vec{v}| \cos \theta$$

$$\vec{u} \cdot \vec{w} = \cos \theta$$

$$|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta \hat{n} = |\vec{u}| |\vec{v}| \sin \theta = \sin \theta$$

5. STATEMENT-1 : If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$.

and

STATEMENT-2 : $(\vec{x} + \vec{y})^2 = |\vec{x}|^2 + |\vec{y}|^2 + 2(\vec{x} \cdot \vec{y})$.

Sol. Answer (1)

$$(\vec{a} + \vec{b} + \vec{c})^2 = a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$\Rightarrow 0 = 3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

6. STATEMENT-1 : If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, then $\vec{a} - \vec{d}$ is perpendicular to $\vec{b} - \vec{c}$.

and

STATEMENT-2 : If $\vec{P}$ and $\vec{Q}$ are perpendicular then $\vec{P} \cdot \vec{Q} = 0$.

Sol. Answer (4)

$$\vec{a} \times \vec{b} = \vec{c} \times \vec{d} \quad \dots (i)$$

$$\vec{a} \times \vec{c} = \vec{b} \times \vec{d} \quad \dots (ii)$$

$$\vec{a} \times \vec{b} - \vec{a} \times \vec{c} = \vec{c} \times \vec{d} - \vec{b} \times \vec{d}$$

$$\Rightarrow \vec{a} \times (\vec{b} - \vec{c}) = (\vec{c} - \vec{b}) \times \vec{d}$$

$$\vec{a} \times (\vec{b} - \vec{c}) + (\vec{b} - \vec{c}) \times \vec{d} = 0$$

$$(\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = 0$$

$$\Rightarrow \vec{a} - \vec{d} \text{ is parallel to } \vec{b} - \vec{c}$$

Hence statement-1 is false, but statement-2 is true.

7. Let the vectors $\overrightarrow{PO}, \overrightarrow{QR}, \overrightarrow{RS}, \overrightarrow{ST}, \overrightarrow{TU}$ and $\overrightarrow{UP}$ represent the sides of a regular hexagon.

[IIT-JEE 2007]

STATEMENT-1 : $\overrightarrow{PQ} \times (\overrightarrow{RS} + \overrightarrow{ST}) \neq \vec{0}$

and

STATEMENT-2 : $\overrightarrow{PQ} \times \overrightarrow{RS} = \vec{0}$ and $\overrightarrow{PQ} \times \overrightarrow{ST} \neq \vec{0}$

Sol. Answer (3)

$\overrightarrow{PQ}$ and $\overrightarrow{RS} + \overrightarrow{ST}$ are not parallel, thus

$$\overrightarrow{PQ} \times (\overrightarrow{RS} + \overrightarrow{ST}) \neq \vec{0}$$

Again $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ are not parallel, thus

$$\overrightarrow{PQ} \times \overrightarrow{RS} \neq \vec{0}$$

$\therefore$ Statement-1 is true, statement-2 is false

SECTION - E

Matrix-Match Type Questions

1. Match the area/volume of column-I and column-II

Column-I

Column-II

- (A) Triangle with vertices $(-\hat{i} + 2\hat{j} + 3\hat{k}), (2\hat{i} - \hat{j} - \hat{k}), (\hat{i} + \hat{j} - \hat{k})$

(p) $\frac{\sqrt{3}}{2}$

- (B) Tetrahedron with vertices $(0, 0, 0), (1, 1, -1), (1, -1, 1), (-1, 1, 1)$

(q) $\frac{2}{3}$

- (C) Tetrahedron with vertices $(-1, 0, 1), (2, -1, 0), (1, 2, 5), (1, 2, 1)$

(r) $\frac{\sqrt{89}}{2}$

- (D) Triangle with vertices $(1, 0, 0), (0, 1, 0), (1, 1, 1)$

(s) $\frac{16}{3}$

Sol. Answer : A(r), B(q), C(s), D(p)

(A) Area of triangle = $|\vec{A}|$

$$\text{Where, } \vec{A} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -3 & -4 \\ -1 & 2 & 0 \end{vmatrix}$$

$$\Rightarrow \vec{A} = \frac{1}{2} |\hat{i}(0+8) - \hat{j}(0-4) + \hat{k}(6-3)|$$

$$\Rightarrow \vec{A} = \frac{1}{2} |8\hat{i} + 4\hat{j} + 3\hat{k}|$$

$$\Rightarrow |\vec{A}| = \frac{\sqrt{89}}{2}$$

$$\begin{aligned}
 \text{(B) Volume} &= \frac{1}{6} \begin{vmatrix} 1-0 & 1-0 & -1-0 \\ 1-1 & -1-1 & 1-(-1) \\ -1-1 & 1-(-1) & 1-1 \end{vmatrix} \\
 &= \frac{1}{6} \begin{vmatrix} 1 & 1 & -1 \\ 0 & -2 & 2 \\ -2 & 2 & 0 \end{vmatrix} \\
 &= \frac{1}{6} [1(0-4) - 1(0+4) - 1(0-4)] \\
 &= \frac{1}{6} [-4-4+4]
 \end{aligned}$$

$$\text{Volume} = \frac{2}{3}$$

$$\text{(C) } v = \frac{1}{6} \begin{vmatrix} 2+1 & -1-0 & 0-1 \\ 1-2 & 2+1 & 5-0 \\ 1-1 & 2-2 & 1-5 \end{vmatrix} = \frac{1}{6} \begin{vmatrix} 3 & -1 & -1 \\ -1 & 3 & 5 \\ 0 & 0 & -4 \end{vmatrix} = \frac{16}{3}$$

$$\text{(D) Area} = |\vec{A}|$$

$$\begin{aligned}
 |\vec{A}| &= \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} \\
 &= \frac{1}{2} [\hat{i}(1-0) - \hat{j}(-1-0) + \hat{k}(0-1)] \\
 &= \frac{1}{2} [\hat{i} + \hat{j} - \hat{k}]
 \end{aligned}$$

$$\Rightarrow |\vec{A}| = \frac{\sqrt{3}}{2}$$

2. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be any three vectors then match the following

Column-I

- (A) $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a})$
 (B) $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a})$
 (C) $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{b})$
 (D) $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{a})$

Column-II

- (p) $[\vec{c} \ \vec{b} \ \vec{a}] \vec{a}$
 (q) $[\vec{c} \ \vec{b} \ \vec{a}] \vec{b}$
 (r) $(\vec{a} \times \vec{b}) \times \vec{c}$
 (s) $(\vec{a} + \vec{b}) \times \vec{c}$

Sol. Answer : A(s), B(r), C(q), D(p)

$$(A) \vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{c} + \vec{b} \times \vec{a} = (\vec{a} + \vec{b}) \times \vec{c}$$

$$(B) \vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} + (\vec{b} \cdot \vec{a})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \times \vec{b}) \times \vec{c}$$

$$(C) (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{b}) = [(\vec{a} \times \vec{b}) \cdot \vec{b}]\vec{c} - [(\vec{a} \times \vec{b}) \cdot \vec{c}]\vec{b} = -[\vec{a} \cdot \vec{b} \cdot \vec{c}]\vec{b} = [\vec{c} \cdot \vec{b} \cdot \vec{a}]\vec{b}$$

$$(D) (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{a}) = [(\vec{a} \times \vec{b}) \cdot \vec{a}]\vec{c} - [(\vec{a} \times \vec{b}) \cdot \vec{c}]\vec{a} = 0 - [\vec{a} \cdot \vec{b} \cdot \vec{c}]\vec{a} = [\vec{c} \cdot \vec{b} \cdot \vec{a}]\vec{a}$$

3. Match the following

Column-I

Column-II

(A) ABCD is a rhombus whose diagonals intersect at P then,

(p) Unit vector

$$\vec{PA} + \vec{PB} + \vec{PC} + \vec{PD} \text{ is}$$

(B) If magnitude each of $\vec{a}, \vec{b}, \vec{c}$ is $\frac{1}{2}$ and $\vec{a}$ is perpendicular to the plane

(q) Zero vector

of $\vec{b}$ and $\vec{c}$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$, then $\vec{a} + \vec{b} + \vec{c}$ is

(C) A point I is the centre of the circle inscribed in a triangle ABC. Then

(r) $\frac{\hat{i} + \hat{j} + \hat{k}}{2\sqrt{3}}$

$$|\vec{BC}||\vec{IA}| + |\vec{CA}||\vec{IB}| + |\vec{AB}||\vec{IC}| \text{ is}$$

(D) If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$

(s) Non-zero vector

then $\vec{b}$ is

(t) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

Sol. Answer A(q), B(p, s), C(q), D(p, s)

(A) Obviously

$$(B) |\vec{a}| = |\vec{b}| = |\vec{c}| = \frac{1}{2}, \quad \vec{b} \cdot \vec{c} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

$$\vec{a} \cdot \vec{b} = 0 = \vec{a} \cdot \vec{c}$$

$$\text{Now } |\vec{a} + \vec{b} + \vec{c}|^2 = 1$$

(C) Let I as origin then

$$\text{Given vector } \frac{a\vec{a} + b\vec{b} + c\vec{c}}{a+b+c} = 0$$

(D) $\vec{a}, \vec{b}, \vec{c}$ are perpendicular to each other

$$\text{as } 0 = [\vec{a} \vec{a} \vec{b}] = \vec{a} \cdot \vec{c}, \quad \vec{a} \cdot \vec{b} = 0 = \vec{a} \cdot \vec{c}$$

$$\text{Now } (\vec{a} \times \vec{b}) \cdot \vec{c} = c^2 = a^2 = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$\text{Then } |\vec{a} \times \vec{b}| = |\vec{c}| \Rightarrow ab = c \Rightarrow b = 1$$

4. Match the statements/expressions given in Column I with the values given in Column II.

[IIT-JEE 2009]

Column I

Column II

(A) Root(s) of the equation $2\sin^2\theta + \sin^22\theta = 2$

(p) $\frac{\pi}{6}$

(B) Points of the discontinuity of the function $f(x) = \left[\frac{6x}{\pi} \right] \cos \left[\frac{3x}{\pi} \right]$,
where $[y]$ denotes the largest integer less than or equal to y

(q) $\frac{\pi}{4}$

(C) Volume of the parallelepiped with its edges represented by the
vectors $\hat{i} + \hat{j}$, $\hat{i} + 2\hat{j}$ and $\hat{i} + \hat{j} + \pi\hat{k}$

(r) $\frac{\pi}{3}$

(D) Angle between vectors $\vec{a}$ and $\vec{b}$ where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit
vectors satisfying $\vec{a} + \vec{b} + \sqrt{3}\vec{c} = \vec{0}$

(s) $\frac{\pi}{2}$

(t) π

Sol. Answer A(q, s), B(p, r, s, t), C(t), D(r)

We have,

(A) $2\sin^2\theta + \sin^22\theta = 2$

$$\Rightarrow 2\sin^2\theta + 4\sin^2\theta\cos^2\theta = 2$$

$$\sin^2\theta + 2\sin^2\theta\cos^2\theta = 1$$

$$\sin^2\theta + 2\sin^2\theta(1 - \sin^2\theta) = 1$$

$$3\sin^2\theta - 2\sin^4\theta = 1$$

$$2\sin^4\theta - 3\sin^2\theta + 1 = 0$$

$$2\sin^4\theta - 2\sin^2\theta - \sin^2\theta + 1 = 0$$

$$2\sin^2\theta(\sin^2\theta - 1) - 1(\sin^2\theta - 1) = 0$$

$$(\sin^2\theta - 1)(2\sin^2\theta - 1) = 0$$

$$\Rightarrow \sin\theta = \pm 1, \sin\theta = \pm \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{2}, -\frac{\pi}{2}, \theta = \frac{\pi}{4}, -\frac{\pi}{4}$$

(B) $f(x) = \left[\frac{6x}{\pi} \right] \cos \left[\frac{3x}{\pi} \right]$

Point of discontinuity are the points where

$$\frac{6x}{\pi} \text{ and } \frac{3x}{\pi} \text{ are integer}$$

$$\frac{6x}{\pi} = l_1 \Rightarrow x = \frac{l_1\pi}{6} = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{6}, \frac{5\pi}{6}, \pi$$

$$\frac{3x}{\pi} = l_2 \Rightarrow x = \frac{l_2\pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}, \pi$$

$$(C) V = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & \pi \end{vmatrix} = \pi$$

$$(D) \vec{a} + \vec{b} = -\vec{c}\sqrt{3}$$

$$\Rightarrow 1+1+2\cos\theta = 3$$

$$\Rightarrow \cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

5. Match the statements given in Column I with the values given in Column II

[IIT-JEE 2011]

Column I

Column II

(A) If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}$, $\vec{b} = -\hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ form a triangle, then internal angle of the triangle between $\vec{a}$ and $\vec{b}$ is

(p) $\frac{\pi}{6}$

(B) If $\int_a^b (f(x) - 3x)dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is

(q) $\frac{2\pi}{3}$

(C) The value of $\frac{\pi^2}{\ln 3} \int_{\frac{7}{6}}^{\frac{5}{6}} \sec(\pi x)dx$ is

(r) $\frac{\pi}{3}$

(D) The maximum value of $\left| \arg\left(\frac{1}{1-z}\right) \right|$ $|z|=1, z \neq 1$ is given by

(s) π

(t) $\frac{\pi}{2}$

Sol. Answer A(r), B(p), C(s), D(t)

(A) We have,

$$\vec{a} = \hat{j} + \sqrt{3}\hat{k}$$

$$\vec{b} = -\hat{j} + \sqrt{3}\hat{k}$$

$$\vec{c} = 2\sqrt{3}\hat{k}$$

We observe that

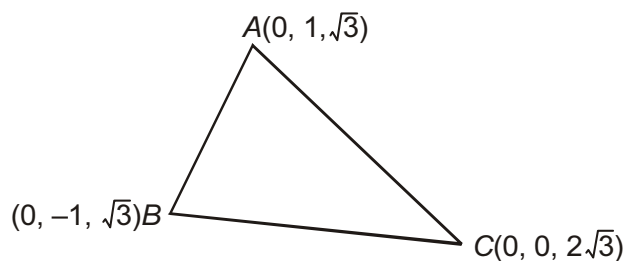
$$\vec{a} + \vec{b} = \vec{c}$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = |\vec{c}|^2$$

$$\Rightarrow 4 + 4 + 8\cos\theta = 12$$

$$\cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$



(B) We have,

$$\int_a^b (f(x) - 3x) dx = a^2 - b^2$$

Keeping a constant and differentiating both sides w.r.t. b , we get

$$f(b) - 3b = -2b$$

$$\Rightarrow f(b) = b$$

$$\Rightarrow f\left(\frac{\pi}{6}\right) = \frac{\pi}{6}$$

(C) Using $\int \sec \pi x dx = \frac{1}{\pi} \ln |\sec \pi x + \tan \pi x|$, we get

$$\frac{\pi^2}{\ln 3} \int_{7/6}^{5/6} \sec \pi x dx = \pi$$

$$(D) \left| \arg\left(\frac{1}{1-z}\right) \right| = |\arg(1) - \arg(1-z)|$$

$$= |\arg(1-z)|$$

But z lies on $|z| = 1$

Hence it is obvious from geometry that $\arg(1-z)$ can be

very-very close to $\frac{\pi}{2}$, if the chosen complex

number is very close to 1.

Let us suppose that $z = \cos\theta + i\sin\theta$

$$\frac{1}{1-z} = \frac{1}{2} \left(1 + i \cot \frac{\theta}{2} \right)$$

Now $-\pi < \theta \leq \pi$, $\theta \neq 0$

$$\arg\left(\frac{1}{1-z}\right) = \frac{\pi}{2} - \frac{\theta}{2}, \quad 0 < \theta \leq \frac{\pi}{2}$$

$$\text{and } \arg\left(\frac{1}{1-z}\right) = \pi - \left(\frac{\pi}{2} - \frac{\theta}{2}\right), \quad -\frac{\pi}{2} < \theta < 0$$

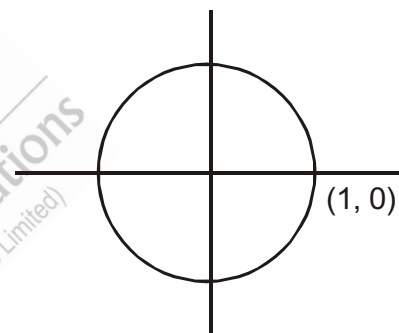
$$= \frac{\pi}{2} + \frac{\theta}{2}, \quad -\frac{\pi}{2} < \theta < 0$$

So in both the cases we can observe that $\arg\left(\frac{1}{1-z}\right)$ can take values which are infinitesimally closer to $\frac{\pi}{2}$,

but given the statement of the problem, this value can't be achieved.

It can be seen that if one takes z to be a complex number which is very-very close to $(1 + 0i)$, the argument

of $\left(\frac{1}{1-z}\right)$ will be very close to $\frac{\pi}{2}$.



6. Match Column I with Column II and select the correct answer using the code given below the lists

[JEE(Advanced) 2013]

Column I

Column II

- (A) Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelepiped determined by vectors $2(\vec{a} \times \vec{b}), 3(\vec{b} \times \vec{c})$ and $(\vec{c} \times \vec{a})$ is (p) 100
- (B) Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelepiped determined by vectors $3(\vec{a} + \vec{b}), (\vec{b} + \vec{c})$ and $2(\vec{c} + \vec{a})$ is (q) 30
- (C) Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is (r) 24
- (D) Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $(\vec{a} + \vec{b})$ and $\vec{a}$ is (s) 60

Codes

A	B	C	D
(1) s	q	r	p
(2) q	r	p	s
(3) r	s	p	q
(4) p	s	r	q

Sol. Answer (3)

(A) Given, $[\vec{a} \ \vec{b} \ \vec{c}] = 2$

$$\begin{aligned} \text{Now, } V &= [2(\vec{a} \times \vec{b}) \ 3(\vec{b} \times \vec{c}) \ \vec{c} \times \vec{a}] \\ &= 6[\vec{a} \ \vec{b} \ \vec{c}]^2 \\ &= 24 \end{aligned}$$

(B) Given, $[\vec{a} \ \vec{b} \ \vec{c}] = 5$

$$\begin{aligned} \text{Now, } V &= [3(\vec{a} + \vec{b}) \ \vec{b} + \vec{c} \ 2(\vec{c} + \vec{a})] \\ &= 6[\vec{a} + \vec{b} \ \vec{b} + \vec{c} \ \vec{c} + \vec{a}] \\ &= 12[\vec{a} \ \vec{b} \ \vec{c}] \\ &= 60 \end{aligned}$$

(C) Given, $|\vec{a} \times \vec{b}| = 40$

$$\begin{aligned}\text{Now, } A &= \frac{1}{2} |(2\vec{a} + 3\vec{b}) \times (\vec{a} - \vec{b})| \\ &= \frac{1}{2} \cdot 5 |\vec{a} + \vec{b}| \\ &= \frac{5}{2} \times 40 \\ &= 100\end{aligned}$$

(D) Given, $|\vec{a} \times \vec{b}| = 30$

$$\begin{aligned}\text{Now, } A &= |(\vec{a} + \vec{b}) \times \vec{a}| \\ &= |\vec{a} + \vec{b}| \\ &= 30\end{aligned}$$

7. Match Column I with Column II and select the correct answer using the code given below the lists

[JEE(Advanced) 2014]

Column I

Column II

(A) Let $y(x) = \cos(3 \cos^{-1} x)$, $x \in [-1, 1]$, $x \neq \pm \frac{\sqrt{3}}{2}$. Then

(p) 1

$$\frac{1}{y(x)} \left\{ (x^2 - 1) \frac{d^2 y(x)}{dx^2} + x \frac{dy(x)}{dx} \right\} \text{ equals}$$

(B) Let $A_1, A_2, \dots, A_n$ ($n > 2$) be the vertices of a regular polygon of n sides with its centre at the origin.

(q) 2

Let $\vec{a}_k$ be the position vector of the point A_k ,

$$k = 1, 2, \dots, n. \text{ If } \left| \sum_{k=1}^{n-1} (\vec{a}_k \times \vec{a}_{k+1}) \right| = \left| \sum_{k=1}^{n-1} (\vec{a}_k \cdot \vec{a}_{k+1}) \right|,$$

then the minimum value of n is

(C) If the normal from the point $P(h, 1)$ on the ellipse

(r) 8

$$\frac{x^2}{6} + \frac{y^2}{3} = 1 \text{ is perpendicular to the line } x + y = 8,$$

then the value of h is

(D) Number of positive solutions satisfying the equation

(s) 9

$$\tan^{-1} \left(\frac{1}{2x+1} \right) + \tan^{-1} \left(\frac{1}{4x+1} \right) = \tan^{-1} \left(\frac{2}{x^2} \right) \text{ is}$$

A B C D

- | | | | |
|-------|---|---|---|
| (1) s | r | q | p |
| (2) q | s | r | p |
| (3) s | r | p | q |
| (4) q | s | p | r |

Sol. Answer (1)

$$(A) \frac{dy}{dx} = -\sin(3 \cos^{-1} x) \cdot \frac{-3}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2) \left(\frac{dy}{dx} \right)^2 = 9(1-\cos^2(3 \cos^{-1} x)) = 9(1-y^2)$$

$$\Rightarrow (1-x^2)^2 \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \cdot (-2x) = -18y \cdot \frac{dy}{dx}$$

$$\Rightarrow (x^2-1) \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right) = 9y$$

$$(B) (n-1) \cdot R^2 \sin\left(\frac{2\pi}{n}\right) = (n-1)R^2 \cos \frac{2\pi}{n}$$

$$\Rightarrow \tan\left(\frac{2\pi}{n}\right) = 1$$

$$\Rightarrow n = 8$$

(C) As, equation of normal is

$$\frac{\sqrt{6}x}{\cos \theta} - \frac{\sqrt{3}y}{\sin \theta} = 3$$

$$\text{Given its slope} = 1 \Rightarrow \tan \theta = \frac{1}{\sqrt{2}}$$

Also (i) passes through $(h, 1)$

$$\text{So, } \frac{\sqrt{6}h}{\left(\frac{\sqrt{2}}{\sqrt{3}}\right)} - \frac{\sqrt{3}}{\left(\frac{1}{\sqrt{3}}\right)} = 3 \Rightarrow h = 2$$



$$(D) \tan^{-1} \left(\frac{\frac{1}{2x+1} + \frac{1}{4x+1}}{1 - \frac{1}{(2x+1)(4x+1)}} \right) = \tan^{-1} \left(\frac{2}{x^2} \right)$$

$$\Rightarrow \frac{6x+2}{8x^2+6x} = \frac{2}{x^2} \Rightarrow \frac{3x+1}{4x+3} = \frac{2}{x} \text{ (where } x \neq 0)$$

$$\Rightarrow 3x^2 - 7x - 6 = 0$$

$$\Rightarrow x = 3, -\frac{2}{3}$$

But $2x+1 > 0$ and $4x+1 > 0$ So, solution are $x = 3$

SECTION - F

Integer Answer Type Questions

1. If $\vec{x}_1, \vec{x}_2, \vec{x}_3$ and $\vec{y}_1, \vec{y}_2, \vec{y}_3$ are two sets of non-coplanar vectors such that

$$\vec{x}_r \cdot \vec{y}_s = \begin{cases} 0, & \text{if } r \neq s \\ 2 & \text{if } r = s \end{cases} \text{ where } r, s = 1, 2, 3$$

What is the value of $[\vec{x}_1 \vec{x}_2 \vec{x}_3][\vec{y}_1 \vec{y}_2 \vec{y}_3]$?

Sol. Answer (8)

$$\vec{x}_1 \cdot \vec{y}_2 = 0, \quad \vec{x}_1 \cdot \vec{y}_3 = 0$$

$$\text{So } \vec{x}_1 \parallel (\vec{y}_2 \times \vec{y}_3)$$

$$\Rightarrow \vec{x}_1 = \lambda (\vec{y}_2 \times \vec{y}_3)$$

$$\Rightarrow \vec{x}_1 \cdot \vec{y}_1 = \lambda [\vec{y}_1 \vec{y}_2 \vec{y}_3]$$

$$\Rightarrow \lambda = \frac{2}{[\vec{y}_1 \vec{y}_2 \vec{y}_3]}$$

$$\text{So } \vec{x}_1 = \frac{2}{[\vec{y}_1 \vec{y}_2 \vec{y}_3]} \cdot (\vec{y}_2 \times \vec{y}_3)$$

Similarly

$$\vec{x}_2 = \frac{2}{[\vec{y}_1 \vec{y}_2 \vec{y}_3]} \cdot (\vec{y}_3 \times \vec{y}_1)$$

$$\vec{x}_3 = \frac{2}{[\vec{y}_1 \vec{y}_2 \vec{y}_3]} \cdot (\vec{y}_1 \times \vec{y}_2)$$

$$\therefore [\vec{x}_1 \vec{x}_2 \vec{x}_3] = \frac{8}{[\vec{y}_1 \vec{y}_2 \vec{y}_3]^3} [\vec{y}_1 \vec{y}_2 \vec{y}_3]^2$$

$$\Rightarrow [\vec{x}_1 \vec{x}_2 \vec{x}_3][\vec{y}_1 \vec{y}_2 \vec{y}_3] = 8$$

2. Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors of which every pair is non collinear, and the vectors $\vec{a} + 3\vec{b}$ and $2\vec{b} + 3\vec{c}$ are collinear with $\vec{c}$ are $\vec{a}$ respectively. If $\vec{b} \cdot \vec{b} = 1$, then find $|2\vec{a} + 9\vec{c}|$.

Sol. Answer (6)

$$\text{Given } \vec{a} + 3\vec{b} = \lambda \vec{c}$$

$$2\vec{b} + 3\vec{c} = \mu \vec{a}$$

$$\Rightarrow 2\vec{b} + 3\vec{c} = \mu(\lambda \vec{c} - 3\vec{b})$$

$$\Rightarrow (2+3\mu)\vec{b} + (3-\mu\lambda)\vec{c} = \vec{0}$$

$$\Rightarrow \mu = -\frac{2}{3}$$

$$\text{Thus, } 2\vec{a} + 6\vec{b} + 9\vec{c} = \vec{0}$$

$$\Rightarrow |2\vec{a} + 9\vec{c}| = 6|\vec{b}| = 6$$

3. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}$, then the value of

$$(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$$

[IIT-JEE 2010]

Sol. Answer (5)

From the given information, it is clear that

$$\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}} \Rightarrow |\vec{a}| = 1, |\vec{b}| = 1, \vec{a} \cdot \vec{b} = 0$$

$$\text{Now, } (2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$$

$$= (2\vec{a} + \vec{b}) \cdot [a^2\vec{b} - (\vec{a} \cdot \vec{b})\vec{a} + 2b^2\vec{a} - 2(\vec{b} \cdot \vec{a})\vec{a}]$$

$$= [2\vec{a} + \vec{b}] \cdot [\vec{b} + 2\vec{a}]$$

$$= 4a^2 + b^2$$

$$= 4 \cdot 1 + 1 = 5 \quad [\text{as } \vec{a} \cdot \vec{b} = 0]$$

4. Let $\vec{a} = -\hat{i} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 2\hat{j} + 3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

[IIT-JEE 2011]

Sol. Answer (9)

We have

$$\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$$

$$\Rightarrow (\vec{r} - \vec{c}) \times \vec{b} = \vec{0}$$

$$\Rightarrow \vec{r} - \vec{c} = \lambda \vec{b}, \lambda \neq 0$$

$$\Rightarrow \vec{r} = \vec{c} + \lambda \vec{b}$$

$$\text{Since } \vec{r} \cdot \vec{a} = 0$$

$$\Rightarrow (\vec{c} + \lambda \vec{b}) \cdot \vec{a} = 0$$

$$\Rightarrow \lambda = 4$$

$$\Rightarrow \vec{r} \cdot \vec{b} = (\vec{c} + 4\vec{b}) \cdot \vec{b} = (-3\hat{i} + 6\hat{j} + 3\hat{k}) \cdot (-\hat{i} + \hat{j}) = 3 + 6 = 9$$

5. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$, then $|2\vec{a} + 5\vec{b} + 5\vec{c}|$ is

[IIT-JEE 2012]

Sol. Answer (3)

$$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$$

$$\Rightarrow 2(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{a}) = 9 \quad \dots(i)$$

$$\Rightarrow 2(3 - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{a}) = 9$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2} \Rightarrow \text{angle between each pair is } 120^\circ.$$

$$\text{Now, } |2\vec{a} + 5\vec{b} + 5\vec{c}|^2 = 4|\vec{a}|^2 + 25|\vec{b}|^2 + 25|\vec{c}|^2 + 20\vec{a} \cdot \vec{b} + 20\vec{a} \cdot \vec{c} + 50\vec{a} \cdot \vec{c}$$

$$= 4 + 25 + 25 - \frac{20}{2} - \frac{20}{2} - \frac{50}{2}$$

$$= 54 - 10 - 10 - 25$$

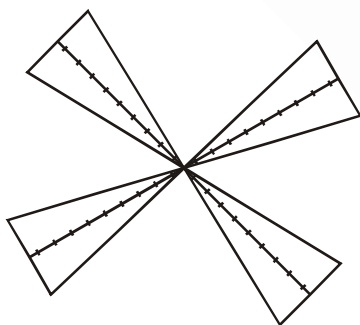
$$= 9$$

$$\therefore |2\vec{a} + 5\vec{b} + 5\vec{c}| = 3$$

6. Consider the set of eight vectors $V = \{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\}$. Three non-coplanar vectors can be chosen from V in 2^p ways. Then p is

[JEE(Advanced) 2013]

Sol. Answer (5)



Total 8 vectors are shown in the figure.

$$\text{Total number of vectors} = {}^8C_3 = 56$$

$$\text{Number of coplaners} = 2 \times (6 \times 2) = 24$$

$$56 - 24 = 32 \approx 2^5$$

7. Let $\vec{a}$, $\vec{b}$, and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$. If

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}, \text{ where } p, q \text{ and } r \text{ are scalars, then the value of } \frac{p^2 + 2q^2 + r^2}{q^2} \text{ is}$$

[JEE(Advanced)-2014]

Sol. Answer (4)

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$$

$$\begin{aligned} \text{Now } [\vec{a} \quad \vec{b} \quad \vec{c}]^2 &= \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} \\ &= 1\left(1 - \frac{1}{4}\right) - \frac{1}{2}\left(\frac{1}{2} - \frac{1}{4}\right) + \frac{1}{2}\left(\frac{1}{4} - \frac{1}{2}\right) \\ &= \frac{3}{4} - \frac{1}{8} - \frac{1}{8} \\ &= \frac{1}{2} \end{aligned}$$

$$\therefore [\vec{a} \quad \vec{b} \quad \vec{c}] = \pm \frac{1}{\sqrt{2}}$$

$$\text{Now, } \vec{a} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c}) = p + \frac{q}{2} + \frac{r}{2}$$

$$\pm \frac{1}{\sqrt{2}} = p + \frac{q}{2} + \frac{r}{2}$$

$$2p + q + r = \pm\sqrt{2} \quad \dots(i)$$

$$\vec{b} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c}) = \frac{p}{2} + q + \frac{r}{2}$$

$$\Rightarrow p + 2q + r = 0 \quad \dots(ii)$$

$$\vec{c} \cdot (\vec{a} \times \vec{b} + \vec{b} \times \vec{c}) = \frac{p}{2} + \frac{q}{2} + r$$

$$p + q + 2r = \pm\sqrt{2} \quad \dots(iii)$$

Now,

$$p = r = -q$$

$$p = r = \pm \frac{1}{\sqrt{2}}, q = \mp \frac{1}{\sqrt{2}}$$

$$\frac{p^2 + 2q^2 + r^2}{q^2} = 4$$

8. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a} \cdot \vec{b} = 0$. For some $x, y \in \mathbb{R}$, let $\vec{c} = x\vec{a} + y\vec{b} + (\vec{a} \times \vec{b})$. If $|\vec{c}| = 2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8 \cos^2 \alpha$ is _____.

[JEE(Advanced)-2018]

Sol. Answer (3)

$$\vec{c} = x\vec{a} + y\vec{b} + (\vec{a} \times \vec{b}) \quad \dots(i)$$

$$\vec{c} \cdot \vec{a} = x + y\vec{b} \cdot \vec{a} + (\vec{a} \times \vec{b}) \cdot \vec{a}$$

$$\Rightarrow x = 2 \cos \alpha$$

$$\text{Also, } \vec{c} \cdot \vec{b} = x\vec{a} \cdot \vec{b} + y|\vec{b}|^2 + (\vec{a} \times \vec{b}) \cdot \vec{b}$$

$$\Rightarrow y = 2 \cos \alpha$$

Squaring (i);

$$|\vec{c}|^2 = x^2 + y^2 + |\vec{a} \times \vec{b}|^2$$

$$\Rightarrow 4 = 8 \cos^2 \alpha + 1$$

$$8 \cos^2 \alpha = 3$$

9. Consider the cube in the first octant with sides OP , OQ and OR of length 1, along the x-axis, y-axis and z-axis, respectively, where $O(0, 0, 0)$ is the origin. Let $S\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ be the centre of the cube and T be the vertex of the cube opposite to the origin O such that S lies on the diagonal OT . If $\vec{p} = \vec{SP}$, $\vec{q} = \vec{SQ}$, $\vec{r} = \vec{SR}$ and $\vec{t} = \vec{ST}$, then the value of $|(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t})|$ is _____.

[JEE(Advanced)-2018]

Sol. Answer (0.50)

$$\vec{p} = \vec{SP} = \frac{1}{2}\hat{i} - \frac{1}{2}\hat{j} - \frac{1}{2}\hat{k} = \frac{1}{2}(\hat{i} - \hat{j} - \hat{k})$$

$$\vec{q} = \vec{SQ} = \frac{1}{2}(-\hat{i} + \hat{j} - \hat{k})$$

$$\vec{r} = \vec{SR} = \frac{1}{2}(-\hat{i} - \hat{j} + \hat{k})$$

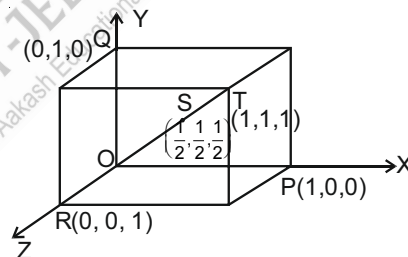
$$\vec{t} = \vec{ST} = \frac{1}{2}(\hat{i} + \hat{j} + \hat{k})$$

$$\vec{p} \times \vec{q} = \frac{1}{4} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ -1 & 1 & -1 \end{vmatrix} = \frac{1}{2}(\hat{i} + \hat{j})$$

$$\vec{r} \times \vec{t} = \frac{1}{4} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \frac{1}{4} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = \frac{1}{2}(-\hat{i} + \hat{j})$$

$$(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t}) = \frac{1}{4}((\hat{i} + \hat{j}) \times (-\hat{i} + \hat{j})) = \frac{1}{4} \times 2\hat{k} = \frac{1}{2}\hat{k}$$

$$|(\vec{p} \times \vec{q}) \times (\vec{r} \times \vec{t})| = \frac{1}{2} = 0.50$$



SECTION - G

Multiple True-False Type Question

1. STATEMENT-1 : If $\vec{a}$ and $\vec{b}$ be two given non zero vectors then any vector $\vec{r}$ coplanar with $\vec{a}$ and $\vec{b}$ can be represented as $\vec{r} = x\vec{a} + y\vec{b}$ where x & y are some scalars.

STATEMENT-2 : If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors, then any vector $\vec{r}$ in space can be expressed as $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$, where x, y, z are some scalars.

STATEMENT-3 : If vectors $\vec{a}$ & $\vec{b}$ represent two sides of a triangle, then $\lambda(\vec{a} + \vec{b})$ (where $\lambda \neq 1$) can represent the vector along third side.

(1) F T T

(2) T T F

(3) F T F

(4) T T T

Sol. Answer (2)

SECTION - H

Aakash Challengers Questions

1. If $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ are given vectors, and $\vec{b}$ is such that $\vec{a} \cdot \vec{b} = 3$ and $\vec{a} \times \vec{b} + \vec{c} = 0$ then $|\vec{b}|^2$ is equal to _____.

Sol. $\vec{a} \times \vec{b} = -\vec{c}$

$$\vec{a} \times (\vec{a} \times \vec{b}) = -\vec{a} \times \vec{c}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} = \vec{c} \times \vec{a}$$

$$\Rightarrow 3\vec{a} - 2\vec{b} = \vec{c} \times \vec{a}$$

$$\Rightarrow \vec{b} = \frac{3\vec{a} - \vec{c} \times \vec{a}}{2}$$

$$\text{and } \vec{c} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = \hat{i}(-1-1) - \hat{j}(-1) + \hat{k}(1) = -2\hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow \vec{b} = \frac{3(\hat{j} - \hat{k}) - (-2\hat{i} + \hat{j} + \hat{k})}{2}$$

$$\Rightarrow \vec{b} = \frac{2\hat{i} + 2\hat{j} - 4\hat{k}}{2} = \hat{i} + \hat{j} - 2\hat{k}$$

$$\Rightarrow |\vec{b}| = \sqrt{6} \Rightarrow |\vec{b}|^2 = 6.$$

2. If $\vec{d} = \lambda(\vec{a} \times \vec{b}) + \mu(\vec{b} \times \vec{c}) + t(\vec{c} \times \vec{a})$, $[\vec{a} \ \vec{b} \ \vec{c}] = \frac{1}{8}$ and $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = 8$ then $\lambda + \mu + t$ equals _____.

Sol. $\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c}) = \mu[abc] + \lambda[abc] + t[abc]$

$$8 = [abc][\mu + \lambda + t]$$

$$8 = \frac{1}{8}[\lambda + \mu + t]$$

$$\Rightarrow \lambda + \mu + t = 64$$

3. If $[\vec{b} \ \vec{c} \ \vec{d}] = 24$ and $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{a} \times \vec{c}) \times (\vec{d} \times \vec{b}) + (\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c}) + k\vec{a} = 0$ then k is equal to _____.

Sol. $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c} \times \vec{d})\vec{b} - (\vec{b} \cdot \vec{c} \times \vec{d})\vec{a}$

$$= [\vec{a} \vec{c} \vec{d}]\vec{b} - [\vec{b} \vec{c} \vec{d}]\vec{a} \quad \dots (i)$$

$$(\vec{a} \times \vec{c}) \times (\vec{a} \times \vec{b}) = (\vec{a} \cdot \vec{d} \times \vec{b})\vec{c} - (\vec{c} \cdot \vec{d} \times \vec{b})\vec{a}$$

$$= [\vec{a} \vec{d} \vec{b}]\vec{c} - [\vec{c} \vec{d} \vec{b}]\vec{a} \quad \dots (ii)$$

$$\text{and } (\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{d} \cdot \vec{c})\vec{b} - (\vec{a} \times \vec{d} \cdot \vec{b})\vec{c}$$

$$= [\vec{a} \vec{d} \vec{c}]\vec{b} - [\vec{a} \vec{d} \vec{b}]\vec{c} \quad \dots (iii)$$

Adding (i), (ii) and (iii), we get

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{a} \times \vec{c}) \times (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c})$$

$$= -[\vec{b} \vec{c} \vec{d}]\vec{a} - [\vec{c} \vec{d} \vec{b}]\vec{a}$$

$$= -2[\vec{b} \vec{c} \vec{d}]\vec{a}$$

$$= -2 \times 24\vec{a} \quad (\because [\vec{b} \vec{c} \vec{d}] = 24)$$

$$= -48\vec{a} = -k\vec{a} \text{ (Given)}$$

On comparison, we get

$$k = 48$$

4. If $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = (\vec{a} \cdot \vec{b} \times \vec{c})^k$ find k .

Sol. $(\vec{a} \times \vec{b}) \cdot [(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})]$

$$= (\vec{a} \times \vec{b}) \cdot [[\vec{b} \vec{c} \vec{a}]\vec{c} - [\vec{b} \vec{c} \vec{c}]\vec{a}]$$

$$= (\vec{a} \times \vec{b}) \cdot [[\vec{b} \vec{c} \vec{a}]\vec{c} - 0]$$

$$= [\vec{a} \vec{b} \vec{c}]^2 = [\vec{a} \vec{b} \vec{c}]^k$$

$$\Rightarrow k = 2$$

5. A plane π passes through the point $(1, 1, 1)$ and is parallel to the vectors $\vec{b} = (1, 0, -1)$ and $\vec{c} = (-1, 1, 0)$. If π meets the axes in A , B , and C , find the volume of the tetrahedron $OABC$.

$$\begin{aligned} \text{Sol. } \vec{n} = \vec{b} \times \vec{c} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{vmatrix} \\ &= \hat{i}(0+1) - \hat{j}(0-1) + \hat{k}(1-0) \\ &= \hat{i} + \hat{j} + \hat{k} \end{aligned}$$

Equation of the plane

$$1(x-1) + 1(y-1) + 1(z-1) = 0$$

$$x + y + z = 3$$

$$\Rightarrow A = (3, 0, 0)$$

$$\text{and } B = (0, 3, 0)$$

$$\text{and } C = (0, 0, 3)$$

$$\Rightarrow \overline{OA} = 3\hat{i}$$

$$\Rightarrow \overline{OB} = 3\hat{j}$$

$$\Rightarrow \overline{OC} = 3\hat{k}$$

$$\text{Volume of tetrahedron} = \frac{1}{6} \begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{vmatrix} = \frac{27}{6} = 4.5$$

6. Solve the simultaneous vector equation of $\vec{x}$ and $\vec{y}$, $\vec{x} + \vec{c} \times \vec{y} = \vec{a}$, $\vec{y} + \vec{c} \times \vec{x} = \vec{b}$, where $\vec{c} \neq 0$.

$$\text{Sol. } \vec{x} + \vec{c} \times \vec{y} = \vec{a} \quad \dots(i)$$

$$\vec{y} + \vec{c} \times \vec{x} = \vec{b} \quad \dots(ii)$$

$$\text{From (ii), } \vec{y} = \vec{b} - \vec{c} \times \vec{x}$$

Using this in (i),

$$\vec{x} + \vec{c} \times (\vec{b} - \vec{c} \times \vec{x}) = \vec{a}$$

$$\Rightarrow \vec{x} + \vec{c} \times \vec{b} - \vec{c} \times (\vec{c} \times \vec{x}) = \vec{a}$$

$$\Rightarrow \vec{x} + \vec{c} \times \vec{b} - (\vec{c} \cdot \vec{x})\vec{c} + (\vec{c} \cdot \vec{c})\vec{x} = \vec{a}$$

$$\text{Let } \vec{c} \cdot \vec{x} = \lambda$$

$$\therefore \bar{x} + \bar{c} \times \bar{b} - \lambda \bar{c} + c^2 \bar{x} = \bar{a}$$

$$\Rightarrow (1 + c^2) \bar{x} = \bar{a} - \bar{c} \times \bar{b} + \lambda \bar{c}$$

$$\Rightarrow (1 + c^2) \bar{x} = \bar{a} + (\bar{b} \times \bar{c}) + \lambda \bar{c} \quad \dots(iii)$$

Since (i) and (ii) are symmetrical in $\bar{x}, \bar{y}$ and $\bar{a}, \bar{b}$ so,

$$(1 + c^2) \bar{y} = \bar{b} + \bar{a} \times \bar{c} + \mu \bar{c} \quad (\bar{c} \cdot \bar{y} = \mu) \quad \dots(iv)$$

Substituting the values found in (iii) and (iv) in the equation (i).

$$\frac{\bar{a} + \bar{b} \times \bar{c} + \lambda \bar{c}}{1 + c^2} + \bar{c} \times \frac{\bar{b} + \bar{a} \times \bar{c} + \mu \bar{c}}{1 + c^2} = \bar{a}$$

$$\Rightarrow \bar{a} + \bar{b} \times \bar{c} + \lambda \bar{c} + \bar{c} \times \bar{b} + \bar{c} \times (\bar{a} \times \bar{c}) + \mu (\bar{c} \times \bar{c}) = \bar{a} + c^2 \bar{a}$$

$$\Rightarrow \bar{b} \times \bar{c} + \lambda \bar{c} - \bar{b} \times \bar{c} + (\bar{c} \cdot \bar{c}) \bar{a} - (\bar{c} \cdot \bar{a}) \bar{c} = c^2 \bar{a}$$

$$\lambda \bar{c} + c^2 \bar{a} - (\bar{c} \cdot \bar{a}) \bar{c} = c^2 \bar{a}$$

$$\lambda \bar{c} - (\bar{c} \cdot \bar{a}) \bar{c} = 0$$

$$\lambda = \bar{c} \cdot \bar{a}$$

Similarly, $\mu = \bar{c} \cdot \bar{b}$

$$\therefore \bar{x} = \frac{\bar{a} + \bar{b} \times \bar{c} + (\bar{c} \cdot \bar{a}) \bar{c}}{1 + c^2}, \bar{y} = \frac{\bar{b} + \bar{a} \times \bar{c} + (\bar{c} \cdot \bar{b}) \bar{c}}{1 + c^2}$$

7. Let $\bar{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$, $\bar{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$, $\bar{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$ be three non-zero vectors such that $\bar{c}$ is a unit vector normal to both the vectors $\bar{a}$ and $\bar{b}$. If angle between $\bar{a}$ and $\bar{b}$ is $\frac{\pi}{6}$. Prove that

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 = \frac{1}{4} (a_1^2 + a_2^2 + a_3^2) (b_1^2 + b_2^2 + b_3^2)$$

$$\text{Sol. } [\bar{a} \bar{b} \bar{c}]^2 = ((\bar{a} \times \bar{b}) \cdot \bar{c})^2 = (|\bar{a}| \cdot |\bar{b}| \cdot \sin \theta \cdot \hat{n} \cdot \bar{c})^2$$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2 = \left(\sqrt{a_1^2 + a_2^2 + a_3^2} \cdot \sqrt{b_1^2 + b_2^2 + b_3^2} \sin \frac{\pi}{6} \cdot |\bar{c}| \cdot \cos \theta \right)^2$$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2 = \frac{1}{4} (a_1^2 + a_2^2 + a_3^2) (b_1^2 + b_2^2 + b_3^2)$$

8. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors and $\vec{u}$ and $\vec{v}$ are any two vectors. Prove that $\vec{u} \times \vec{v} = \frac{1}{[\vec{a} \ \vec{b} \ \vec{c}]} \begin{vmatrix} \vec{u} \cdot \vec{a} & \vec{v} \cdot \vec{a} & \vec{a} \\ \vec{u} \cdot \vec{b} & \vec{v} \cdot \vec{b} & \vec{b} \\ \vec{u} \cdot \vec{c} & \vec{v} \cdot \vec{c} & \vec{c} \end{vmatrix}$.

Sol. Since $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar vectors, can express $\vec{u} \times \vec{v}$ in the form $k_1\vec{a} + k_2\vec{b} + k_3\vec{c}$, where k_1, k_2, k_3 are scalars.

$$\text{i.e., } \vec{u} \times \vec{v} = k_1\vec{a} + k_2\vec{b} + k_3\vec{c}$$

$$(\vec{u} \times \vec{v}) \cdot (\vec{b} \times \vec{c}) = k_1\vec{a} \cdot (\vec{b} \times \vec{c}) + k_2\vec{b} \cdot (\vec{b} \times \vec{c}) + k_3\vec{c} \cdot (\vec{b} \times \vec{c})$$

$$= k_1[\vec{a} \ \vec{b} \ \vec{c}] + 0 + 0$$

$$= k_1[\vec{a} \ \vec{b} \ \vec{c}]$$

$$\Rightarrow \vec{u} \cdot [\vec{v} \times (\vec{b} \times \vec{c})] = k_1[\vec{a} \ \vec{b} \ \vec{c}]$$

$$\Rightarrow \vec{u} \cdot \{(\vec{v} \cdot \vec{c})\vec{b} - (\vec{v} \cdot \vec{b})\vec{c}\} = k_1[\vec{a} \ \vec{b} \ \vec{c}]$$

$$\Rightarrow (\vec{v} \cdot \vec{c})(\vec{u} \cdot \vec{b}) - (\vec{u} \cdot \vec{c})(\vec{v} \cdot \vec{b}) = k_1[\vec{a} \ \vec{b} \ \vec{c}]$$

$$\Rightarrow k_1 = \frac{1}{[\vec{a} \ \vec{b} \ \vec{c}]} \begin{vmatrix} \vec{u} \cdot \vec{b} & \vec{v} \cdot \vec{b} \\ \vec{u} \cdot \vec{c} & \vec{v} \cdot \vec{c} \end{vmatrix}$$

Similarly, taking dot product of (1) with $(\vec{c} \times \vec{a})$ and $(\vec{a} \times \vec{b})$ respectively we get,

$$k_2 = \frac{1}{[\vec{b} \ \vec{c} \ \vec{a}]} \begin{vmatrix} \vec{u} \cdot \vec{c} & \vec{v} \cdot \vec{c} \\ \vec{u} \cdot \vec{a} & \vec{v} \cdot \vec{a} \end{vmatrix}$$

$$k_3 = \frac{1}{[\vec{c} \ \vec{a} \ \vec{b}]} \begin{vmatrix} \vec{u} \cdot \vec{a} & \vec{v} \cdot \vec{a} \\ \vec{u} \cdot \vec{b} & \vec{v} \cdot \vec{b} \end{vmatrix}$$

$$\text{Since } [\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$$

$$\vec{u} \times \vec{v} = \frac{1}{[\vec{a} \ \vec{b} \ \vec{c}]} \left\{ \begin{vmatrix} \vec{u} \cdot \vec{b} & \vec{v} \cdot \vec{b} \\ \vec{u} \cdot \vec{c} & \vec{v} \cdot \vec{c} \end{vmatrix} \cdot \vec{a} + \begin{vmatrix} \vec{u} \cdot \vec{c} & \vec{v} \cdot \vec{c} \\ \vec{u} \cdot \vec{a} & \vec{v} \cdot \vec{a} \end{vmatrix} \cdot \vec{b} + \begin{vmatrix} \vec{u} \cdot \vec{a} & \vec{v} \cdot \vec{a} \\ \vec{u} \cdot \vec{b} & \vec{v} \cdot \vec{b} \end{vmatrix} \cdot \vec{c} \right\}$$

$$= \frac{1}{[\vec{a} \ \vec{b} \ \vec{c}]} \begin{vmatrix} \vec{u} \cdot \vec{a} & \vec{v} \cdot \vec{a} & \vec{a} \\ \vec{u} \cdot \vec{b} & \vec{v} \cdot \vec{b} & \vec{b} \\ \vec{u} \cdot \vec{c} & \vec{v} \cdot \vec{c} & \vec{c} \end{vmatrix}$$

9. If $\begin{vmatrix} p & p^2 & 1+p^3 \\ q & q^2 & 1+q^3 \\ r & r^2 & 1+r^3 \end{vmatrix} = 0$ and the volume of parallelopiped formed by the vectors $\vec{A} = \hat{i} + p\hat{j} + p^2\hat{k}$, $\vec{B} = \hat{i} + q\hat{j} + q^2\hat{k}$ and $\vec{C} = \hat{i} + r\hat{j} + r^2\hat{k}$ is 5. Find pqr .

Sol. $V = [\vec{A} \ \vec{B} \ \vec{C}] = \begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix} = 5 \quad \dots(i)$

Given, $\begin{vmatrix} p & p^2 & 1+p^3 \\ q & q^2 & 1+q^3 \\ r & r^2 & 1+r^3 \end{vmatrix} = 0$

$$\begin{vmatrix} p & p^2 & 1 \\ q & q^2 & 1 \\ r & r^2 & 1 \end{vmatrix} + \begin{vmatrix} p & p^2 & p^3 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix} + pqr \begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix} = 0$$

$$(pqr + 1)(5) = 0$$

$$\Rightarrow pqr = -1$$

10. Let $\vec{a} = 3\hat{i} + 4\hat{j}$, $\vec{b} = 4\hat{j} + 3\hat{k}$. How many vectors $\vec{c}$ are there so that $\vec{a}, \vec{b}, \vec{c}$ are when taken pairwise are equally inclined and $|\vec{a}| = |\vec{b}| = |\vec{c}|$.

Sol. Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$

$$|\vec{a}| = |\vec{b}| = |\vec{c}| \Rightarrow x^2 + y^2 + z^2 = 25 \quad \dots(1)$$

Also, $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{b} \cdot \vec{c}}{|\vec{b}||\vec{c}|} = \frac{\vec{c} \cdot \vec{a}}{|\vec{c}||\vec{a}|}$

$$\Rightarrow 16 = 4y + 3z - 3x + 4y$$

$$\Rightarrow x = z \text{ and } 3x + 4y = 16$$

$$\Rightarrow 41y^2 - 256y + 287 = 0$$

as $D > 0 \Rightarrow y$ has two real and distinct values.

$\Rightarrow$ Two vectors $\vec{c}$ satisfying above condition is possible.

